

INDEX**GitHub Link for all code:**https://github.com/Kevin-Maniar/06Aug_Kevin_SE/tree/main/C%20_%20Assignment%20_%20Code

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Module – 2 | Session – 1**Q – 1**

Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).

Program: -

- ⇒ Program is a set of instructions written for a computer or mobile device to perform a specific task. Program tells a device what to do and how to do it.
- ⇒ For ex. Paytm – Allow user to make payments, receive payments, pay bills, and check for balance.

Q – 2

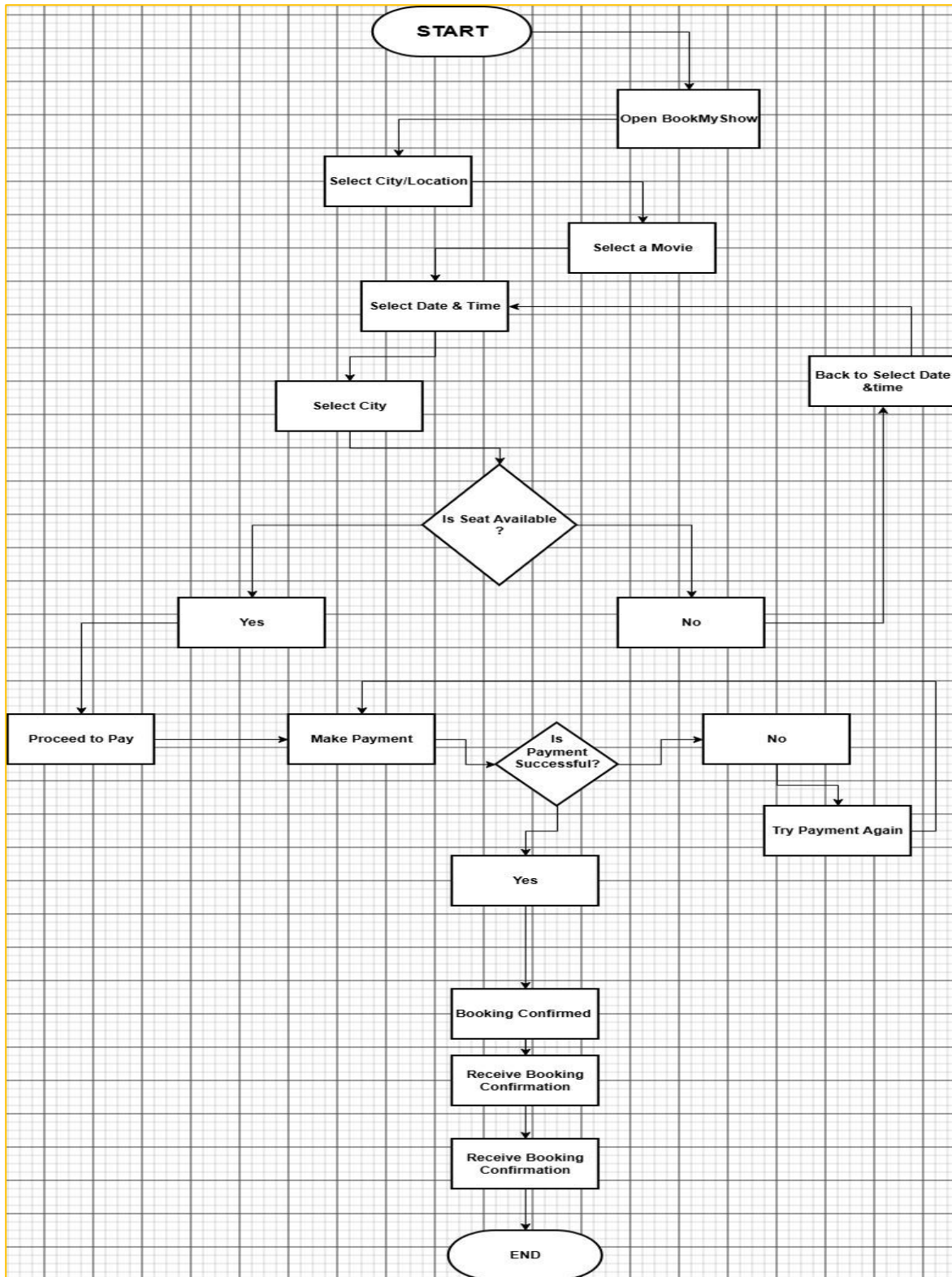
List three Programming languages you have heard of, and for each mention whether it is usually compiled, interpreted, or can be both.

Programming Languages: Java, Python, JavaScript, C++

- ⇒ Java: Compiled
- ⇒ Python: Interpreted
- ⇒ JavaScript: Both Compiled and Interpreted
- ⇒ C++: Compiled

Q – 3

Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on BookMyShow, starting from opening the app to receiving the booking confirmation.



Q – 4

Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.

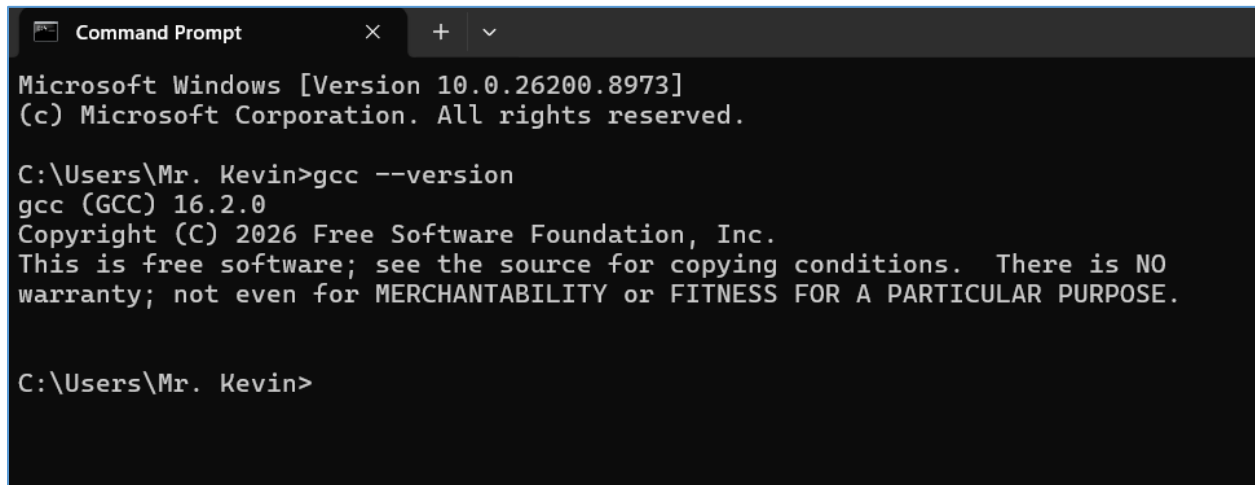
⇒ Algorithm: Ordering food from swiggy

1. Start
2. Open the Swiggy App
3. Log in or Signup
4. Search for the Restaurant or select from the list
5. Select the food items you want to order
6. Add the selected items into the cart
7. Open the cart
8. Enter delivery location and confirm
9. Place order
10. Select the Payment method
11. Complete payment
12. If Payment is Successful, order is confirmed
13. Receive the order confirmation
14. End

Session – 2

Q – 1

Install GCC on your computer and verify the installation by running `gcc --version` in your terminal or command prompt.



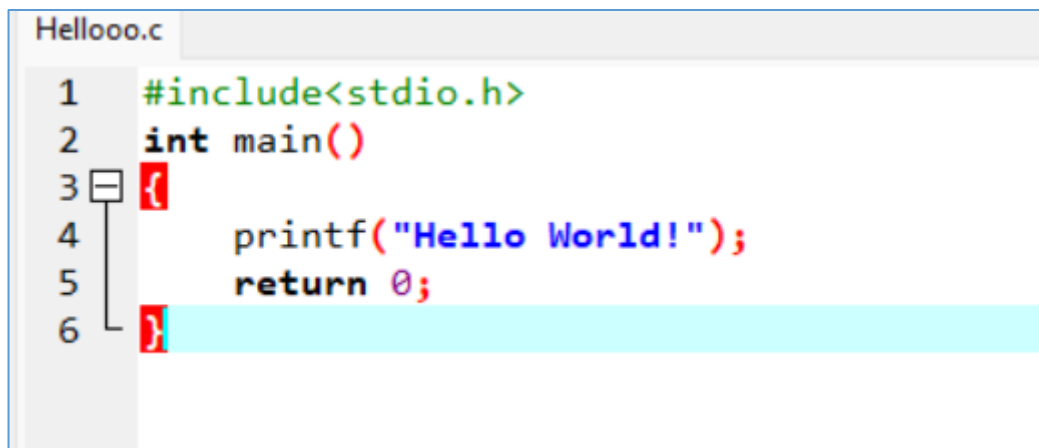
```
Command Prompt
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mr. Kevin>gcc --version
gcc (GCC) 16.2.0
Copyright (C) 2026 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

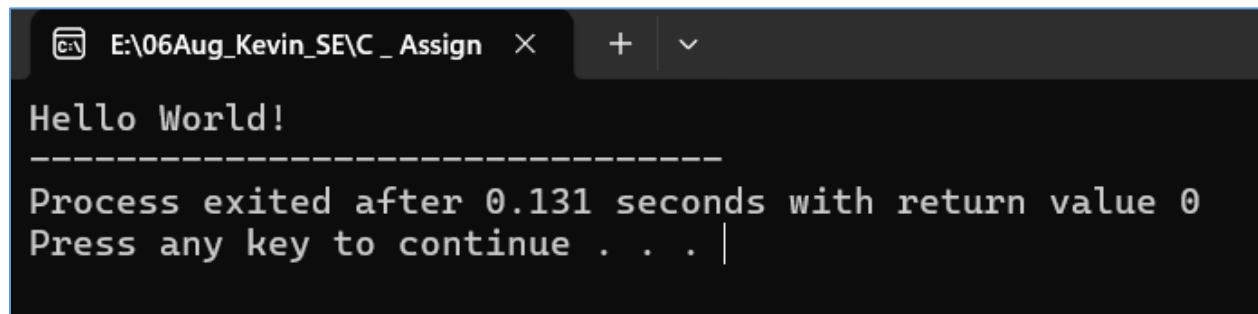
C:\Users\Mr. Kevin>
```

Q – 2

Open VS Code or DevC++ and create a new file named `hello.c`. Write a C program that prints 'Hello, Instagram World!' to the console.



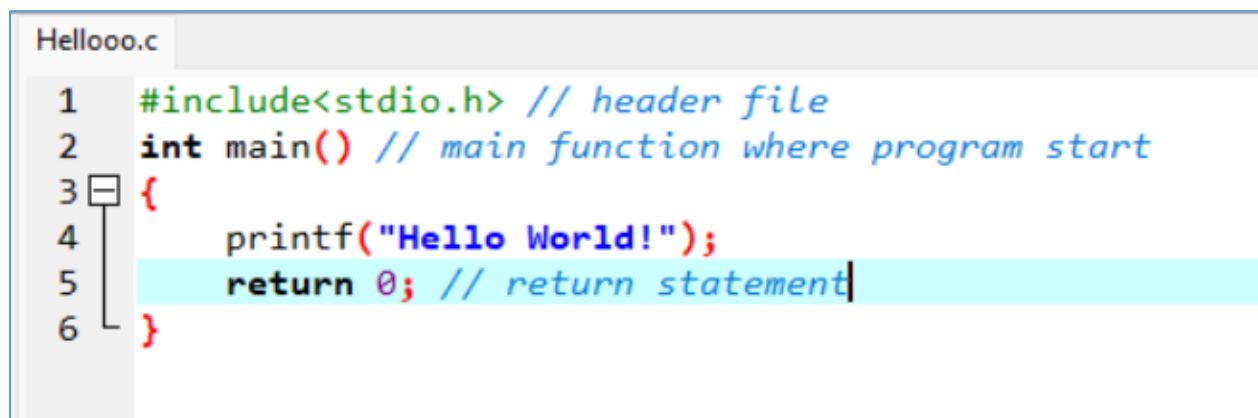
```
Hellooo.c
1  #include<stdio.h>
2  int main()
3  {
4      printf("Hello World!");
5      return 0;
6  }
```



```
E:\06Aug_Kevin_SE\C_Assign × + v
Hello World!
-----
Process exited after 0.131 seconds with return value 0
Press any key to continue . . . |
```

Q-3

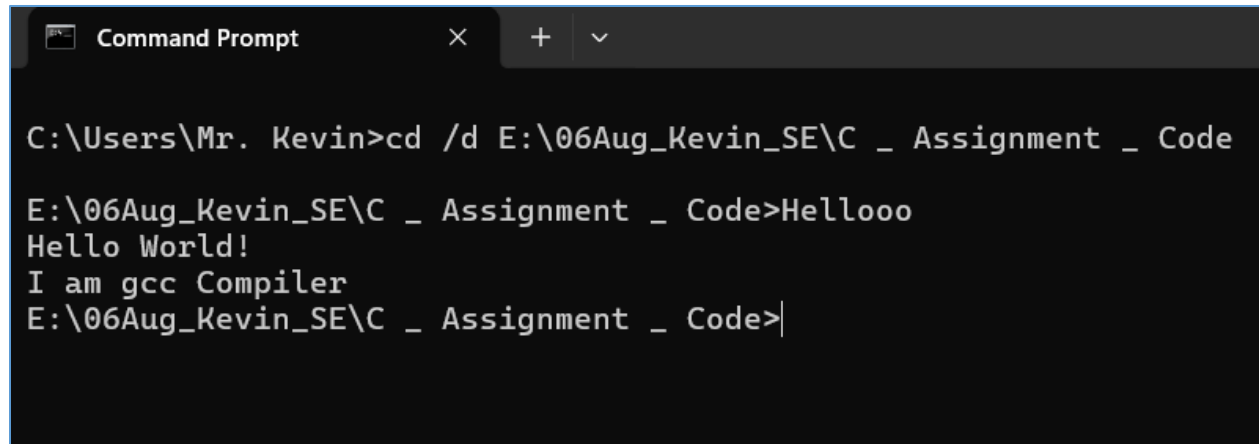
Identify and label the header, main function, and return statement in your hello.c program using comments.



```
Hellooo.c
1  #include<stdio.h> // header file
2  int main() // main function where program start
3  {
4      printf("Hello World!");
5      return 0; // return statement
6  }
```

Q – 4

Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.



```
Command Prompt
C:\Users\Mr. Kevin>cd /d E:\06Aug_Kevin_SE\C _ Assignment _ Code
E:\06Aug_Kevin_SE\C _ Assignment _ Code>Hellooo
Hello World!
I am gcc Compiler
E:\06Aug_Kevin_SE\C _ Assignment _ Code>|
```

Session: 3

Q – 1 Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

The screenshot shows a C program in a code editor and its execution output in a terminal window.

```

S3T1_FP.c
1  #include <stdio.h>
2
3  #define get_type(X) _Generic((X), \
4      float: "float", \
5      double: "double", \
6      char*: "string", \
7      default: "unknown type")
8
9  int main()
10 {
11     char productName[] = "Samsung Galaxy S24";
12     float price = 59999.99;
13     double rating = 4.5;
14
15     printf("Product Name: %s\n", productName);
16     printf("Data Type: %s\n", get_type(productName));
17
18     printf("Price: %.2f\n", price);
19     printf("Data Type: %s\n", get_type(price));
20
21     printf("Rating: %.11f\n", rating);
22     printf("Data Type: %s\n", get_type(rating));
23
24     return 0;
25 }
    
```

The terminal output shows the following results:

```

E:\06Aug_Kevin_SE\C_Assign
Product Name: Samsung Galaxy S24
Data Type: string

Price: 59999.99
Data Type: float

Rating: 4.5
Data Type: double

-----
Process exited after 0.06464 seconds with return value 0
Press any key to continue . . . |
    
```

Q – 2

Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.

The screenshot shows a C++ program in a code editor and its execution output in a terminal window.

```

S3T1_FP.c  S3T2_Const.cpp
1  #include<stdio.h>
2  int main ()
3  {
4      const float GST_RATE = 0.18; //Gst rate 18%
5      float base_price ;
6
7      printf("Enter Base price of food:");
8      scanf("%f",&base_price);
9
10     double gst = GST_RATE * base_price ;
11     float final_price = base_price + gst;
12
13     printf("final price is :%.2f",final_price);
14     return 0;
15 }
16
    
```

The terminal output shows the following results:

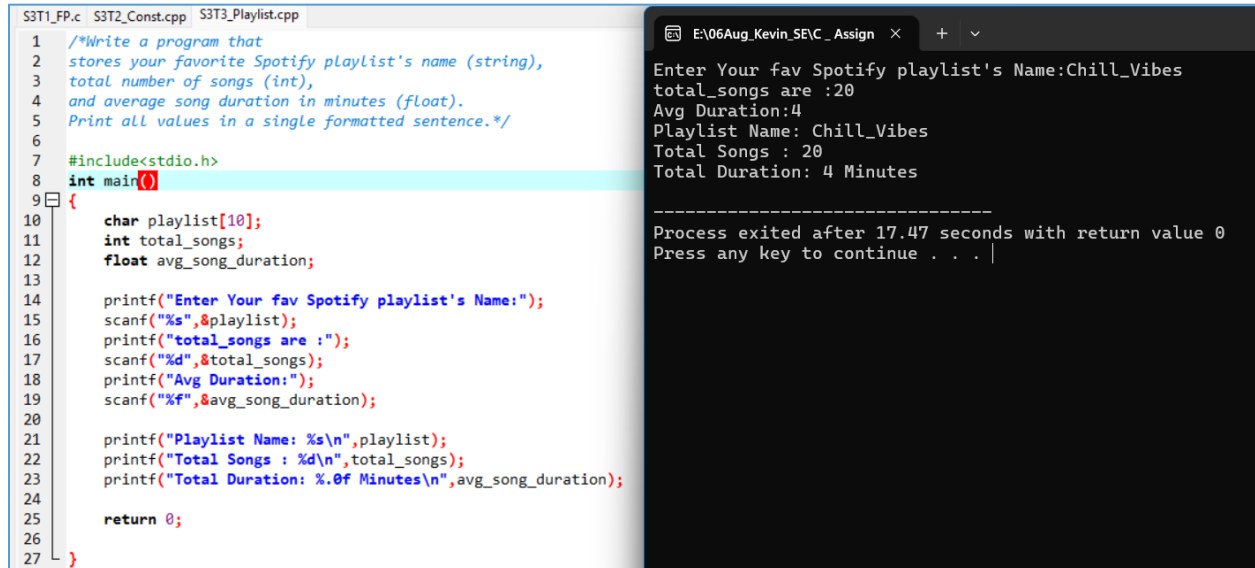
```

E:\06Aug_Kevin_SE\C_Assign
Enter Base price of food:1200
final price is :1416.00

-----
Process exited after 5.663 seconds with return value 0
Press any key to continue . . . |
    
```

Q – 3

Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.



The image shows a C++ program in a code editor and its execution output in a terminal window. The code defines variables for a playlist name, total songs, and average song duration, then prompts the user for input and prints the results in a formatted sentence.

```
S3T1_FP.c S3T2_Const.cpp S3T3_Playlist.cpp
1  /*Write a program that
2  stores your favorite Spotify playlist's name (string),
3  total number of songs (int),
4  and average song duration in minutes (float).
5  Print all values in a single formatted sentence.*/
6
7  #include<stdio.h>
8  int main()
9  {
10     char playlist[10];
11     int total_songs;
12     float avg_song_duration;
13
14     printf("Enter Your fav Spotify playlist's Name:");
15     scanf("%s",&playlist);
16     printf("total_songs are :");
17     scanf("%d",&total_songs);
18     printf("Avg Duration:");
19     scanf("%f",&avg_song_duration);
20
21     printf("Playlist Name: %s\n",playlist);
22     printf("Total Songs : %d\n",total_songs);
23     printf("Total Duration: %.0f Minutes\n",avg_song_duration);
24
25     return 0;
26
27 }
```

Output:

```
E:\06Aug_Kevin_SE\C_Assign x + v
Enter Your fav Spotify playlist's Name:Chill_Vibes
total_songs are :20
Avg Duration:4
Playlist Name: Chill_Vibes
Total Songs : 20
Total Duration: 4 Minutes

-----
Process exited after 17.47 seconds with return value 0
Press any key to continue . . . |
```

Q – 4

Identify and correct the invalid variable names from this list:

1stPlayer, player_age, \$score, total-marks, userName.

- ⇒ Invalid Variable names are : 1stPlayer , \$score , total-marks
- ⇒ Correct Version : first_player , score , total_marks

Session: 4

Que – 1

Create a simple JavaScript function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity` and returns the total bill amount using arithmetic operators.

```

S3T3_Playlist.cpp S4T1_Arithmetic_operator.cpp
1  /*Create a simple JavaScript function called
2  calculateTotal that takes two numbers:
3  itemPrice and quantity, and returns the
4  total bill amount using arithmetic operators.*/
5  #include<stdio.h>
6  int main()
7  {
8
9
10     float itemPrice;
11     int Quantity;
12     float Cal_Total;
13
14     printf("Enter Item Price:");
15     scanf("%f",&itemPrice);
16     printf("Enter Quantity:");
17     scanf("%d",&Quantity);
18
19     Cal_Total = itemPrice * Quantity;
20     printf("Total Bill Amount is: %.2f INR",Cal_Total);
21
22     return 0;
23 }

```

E:\06Aug_Kevin_SE\C_Assign x + v

```

Enter Item Price:120
Enter Quantity:3
Total Bill Amount is: 360.00 INR
-----
Process exited after 2.473 seconds with return value 0
Press any key to continue . . . |

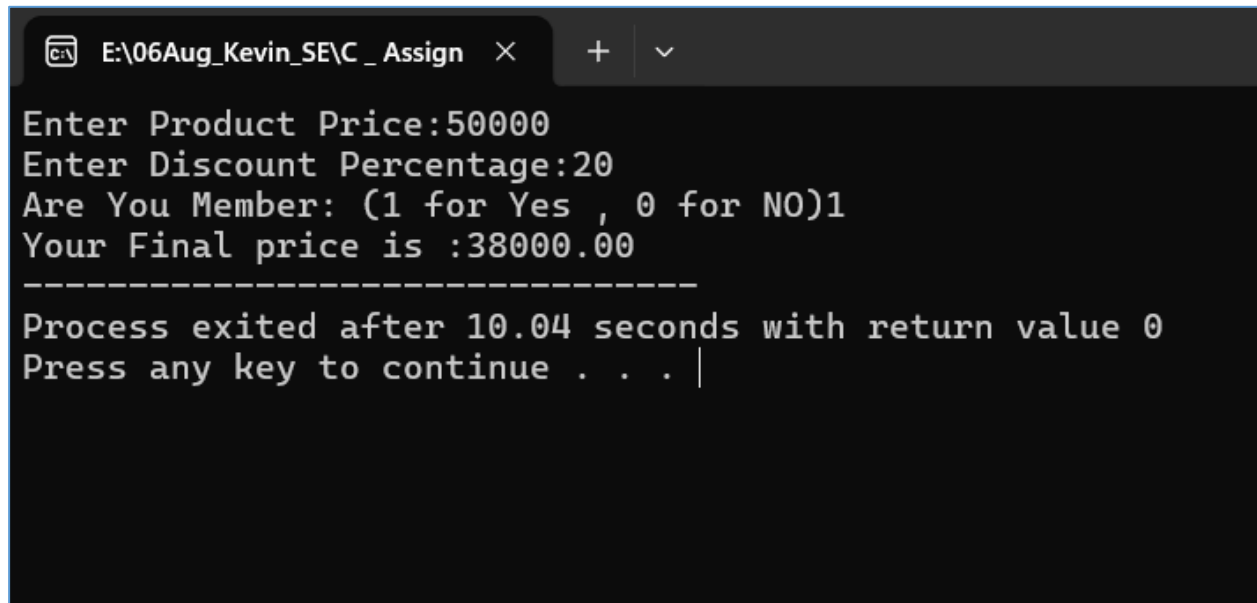
```

Q – 2 Build a Flipkart-style discount calculator given product price, discount percentage, and a boolean `isMember`, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if `isMember` is true).

```

S4T2_Discount_Calculator.cpp
1  #include<stdio.h>
2  int main()
3  {
4      /*Build a Flipkart-style discount calculator:
5      given product price,
6      discount percentage,
7      and a boolean isMember,
8      use arithmetic and logical
9      operators to calculate the final price
10     (apply an extra 5% off if isMember is true). */
11
12     float product_price, discount_percentage;
13     float final_price, discount_price;
14     int isMember;
15
16     printf("Enter Product Price:");
17     scanf("%f",&product_price);
18
19     printf("Enter Discount Percentage:");
20     scanf("%f",&discount_percentage);
21
22     printf("Are You Member: (1 for Yes , 0 for NO)");
23     scanf("%d",&isMember);
24
25     // Count Normal price
26     discount_price = product_price * (discount_percentage/100);
27     //Price After Discount
28     final_price = product_price - discount_price;
29     //Extra 5% discount
30
31     if (isMember == 1)
32     {
33         final_price = final_price -(final_price *5/100);
34         printf("Your Final price is :%.2f",final_price);
35     }
36     else
37     {
38         printf("Final Price = %.2f$",final_price);
39     }
40
41 }

```



```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ∨  
Enter Product Price:50000  
Enter Discount Percentage:20  
Are You Member: (1 for Yes , 0 for NO)1  
Your Final price is :38000.00  
-----  
Process exited after 10.04 seconds with return value 0  
Press any key to continue . . . |
```

Q – 3

Write a function `isEligibleForOffer` that takes a user's age and total order value and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

```
S4T2_Discount_Calculator.cpp S4T3_Relational_logical_oper.cpp
1  /*Write a function isEligibleForOffer
2  that takes a user's age and total order value,
3  and returns true if the user is 18 or older AND
4  the order value is above 500,
5  otherwise false.*/
6
7  #include<stdio.h>
8
9  bool isEligibleForOffer (int age,float order_value)
10 {
11     if(age>=18 && order_value>500)
12     {
13         return true;
14     }
15     else
16     {
17         return false;
18     }
19 }
20
21 int main()
22 {
23     int age;float order_value;
24     printf("Enter Your Age:-");
25     scanf("%d",&age);
26
27     printf("Enter Your Total Order Value:-");
28     scanf("%f",&order_value);
29
30     if(isEligibleForOffer(age,order_value))
31     {
32         printf("YOU ARE ELIGIBLE FOR OFFER");
33     }
34     else{
35         printf("YOU ARE NOT ELIGIBLE FOR OFFER!");
36     }
37     return 0;
38 }
```

```
E:\06Aug_Kevin_SE\C_Assign x + v
Enter Your Age:-18
Enter Your Total Order Value:-5000
YOU ARE ELIGIBLE FOR OFFER
-----
Process exited after 5.618 seconds with return value 0
Press any key to continue . . . |
```

Q – 4

Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

```
S4T4_.cpp
1  /* Given three variables: likes, comments, and shares (all numbers),
2     write code to check if a post is 'trending' on
3     Instagram (at least 1000 likes OR more than
4     200 comments AND at least 50 shares).
5     Print the result. */
6
7  #include<stdio.h>
8  main ()
9  {
10     int likes,comments,share;
11     printf("Enter Total Number of Likes:");
12     scanf("%d",&likes);
13     printf("Enter Total Number of Comments:");
14     scanf("%d",&comments);
15     printf("Enter Total Number Share:");
16     scanf("%d",&share);
17     printf("-----\n");
18     printf("Liked By %d Fan \ncommented %d Fan \nShare By %d Fan\n",likes,comments,share);
19
20     if(likes>=1000 || comments>200 && share>=50)
21     {
22         printf("-----\n");
23         printf("Result: Your Post is Trending! Congratulations");
24     }
25     else
26     {
27         printf("-----\n");
28         printf("Result: Your post is not Trending!!!");
29     }
30 }
```

```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ▾
Enter Total Number of Likes:300
Enter Total Number of Comments:400
Enter Total Number Share:51
-----
Liked By 300 Fan
commented 400 Fan
Share By 51 Fan
-----
Result: Your Post is Trending! Congratulations
-----
Process exited after 29.73 seconds with return value 0
Press any key to continue . . . |
```

Q – 5

Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

```
S4T4_.cpp S4T5_.cpp
1      /* Write a code snippet that
2      demonstrates the difference between pre-increment
3      (++count) and post-increment (count++)
4      by logging the values before and after using both on a
5      followerCount variable. */
6      #include<stdio.h>
7      int main()
8      {
9          int follower = 100;
10
11         printf("Pre Increment Follower ---> %d\n",++follower);
12         printf("Pre Increment Follower ---> %d\n",++follower);
13         printf("Pre Increment Follower ---> %d\n",++follower);
14
15         printf("Post Increment Follower ---> %d\n",follower++);
16         printf("Post Increment Follower ---> %d\n",follower++);
17         printf("Post Increment Follower ---> %d\n",follower++);
18
19         /* follower++ takes 100 first and then add +1
20         ++follower add +1 in exist follower number
21
22         pre increment give result by +1 and
23         post increment give result by using old value first and then +1*/
24         return 0;
25     }
```

```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ∨
Post Increment Follower ---> 100
Post Increment Follower ---> 101
Post Increment Follower ---> 102
Pre Increment Follower ---> 104
Pre Increment Follower ---> 105
Pre Increment Follower ---> 106

-----
Process exited after 0.444 seconds with return value 0
Press any key to continue . . . |
```

Session – 5

Que – 1

Create a simple IPL Fan Bot that takes your favorite IPL team name as input and uses if-else-if statements to print a unique cheer message for each team (e.g., 'Go Mumbai Indians!', 'Chennai Super Kings for the win!'). If the team is not recognized, print 'Team not found!'

```
#include <stdio.h>
#include <string.h>

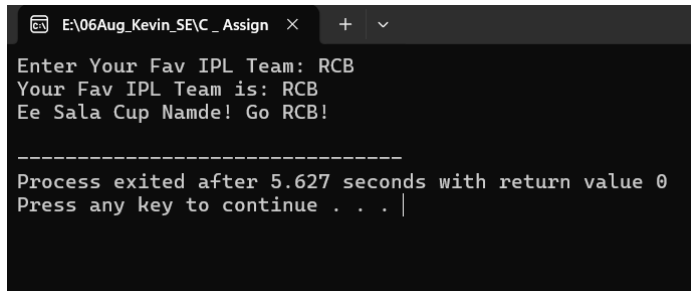
int main() {

    char IPL_team[50];

    printf("Enter Your Fav IPL Team: ");
    scanf("%s", IPL_team);
    printf("Your Fav IPL Team is: %s\n", IPL_team);

    if (strcmp(IPL_team, "MI" ) == 0) {
        printf("Go Mumbai Indians!\n");
    }
    else if (strcmp(IPL_team , "GT") == 0) {
        printf("Aava De!\n");
    }
    else if (strcmp(IPL_team, "CSK") == 0) {
        printf("Chennai Super Kings for the win!\n");
    }
    else if (strcmp(IPL_team, "RCB") == 0) {
        printf("Ee Sala Cup Namde! Go RCB!\n");
    }
    else if (strcmp(IPL_team, "KKR") == 0) {
        printf("Korbo Lorbo Jeetbo Re! Go KKR!\n");
    }
    else {
        printf("Team Not Found!\n");
    }
    return 0;
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SF\C_Assign' and standard window controls. The command prompt displays the following text: 'Enter Your Fav IPL Team: RCB', 'Your Fav IPL Team is: RCB', and 'Ee Sala Cup Namde! Go RCB!'. Below this, a separator line of dashes is shown, followed by the message 'Process exited after 5.627 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

```
E:\06Aug_Kevin_SF\C_Assign > Enter Your Fav IPL Team: RCB
Your Fav IPL Team is: RCB
Ee Sala Cup Namde! Go RCB!

-----
Process exited after 5.627 seconds with return value 0
Press any key to continue . . . |
```

Que – 2

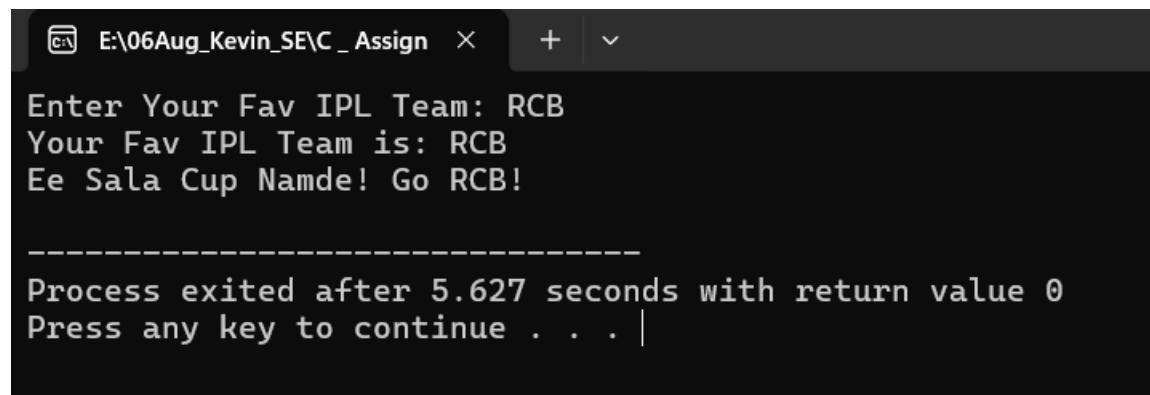
Build a Zomato-style food suggestion tool: take the user's preferred meal time ('breakfast', 'lunch', 'dinner', or 'snack') and use a switch-case statement to suggest a popular dish for that time. If the input doesn't match any meal, suggest 'Try some fruits!'

```
#include <stdio.h>
int main()
{
    int choice;
    printf("--- Food Suggestion Tool ---\n");
    printf("-----\n");
    printf("1..BreakFast\n2..Lunch\n3..Dinner\n4..Snack\n");
    printf("\n");
    printf("Enter Your Choice:");
    scanf("%d",&choice);
    printf("\n");

    switch (choice)
    {
        case 1:
            printf("Oh! BreakFast Time\n");
            printf("You can have\n");
            printf("1...OatMeal\n");
            printf("2...Eggs\n");
            printf("3...Poha\n");
            printf("4...Avocado\n");
            printf("5...Toast\n");
            printf("6...Milk\n");
            break;
        case 2:
            printf("Oh! Lunch Time\n");
            printf("You can have\n");
            printf("1...Roti With Dal\n");
            printf("2...Punjabi Dish\n");
            printf("3...Gujarati Thali\n");
            printf("4...Paneer Sabji\n");
            break;
        case 3:
            printf("Oh! Dinner Time\n");
            printf("You can have\n");
            printf("1..Vegetable Khichadi\n");
            printf("2..Snadwitch\n");
            printf("3..Chhole Bhature\n");
            printf("4..Masala Dosa\n");
```

```
        printf("5..Idli\n");
    break;
    case 4:
        printf("Oh! Snack Time\n");
        printf("You can have\n");
        printf("1..Mixed Nuts & Almonds\n");
        printf("2..Roasted Makhna\n");
        printf("3..SoyaChunks\n");
        printf("4..Wafors\n");
        printf("5..Sev Mummra\n");
    break;
    default:
        printf("Have Some Fruits\n");
    break;
}
return 0;
}
```

Output: -



```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ▾

Enter Your Fav IPL Team: RCB
Your Fav IPL Team is: RCB
Ee Sala Cup Namde! Go RCB!

-----
Process exited after 5.627 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

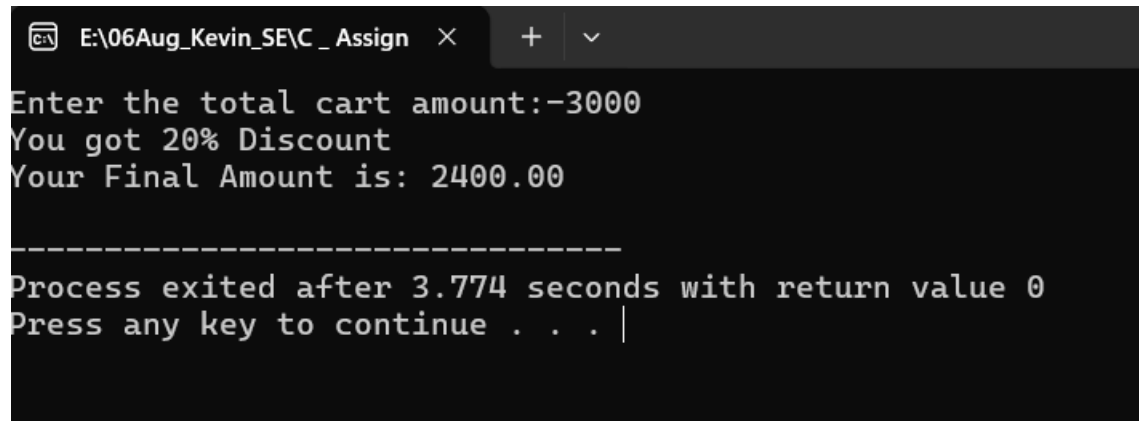
Create a Flipkart discount calculator that asks the user for the total cart amount. Use nested if statements to check: if amount > 2000, apply 20% discount; else if amount > 1000, apply 10% discount; else, no discount. Print the final amount to pay.

Hint: Use nested ifs to check each discount slab.

```
#include<stdio.h>
int main()
{
    float cart_amount;
    float final_price;
    printf("Enter the total cart amount:-");
    scanf("%f",&cart_amount);

    if(cart_amount>2000)
    {
        final_price = cart_amount*0.80;
        printf("You got 20%% Discount\n");
        printf("Your Final Amount is: %.2f\n",final_price);
    }
    else
    {
        if(cart_amount>1000)
        {
            final_price = cart_amount*0.90;
            printf("You got 10%% Discount\n");
            printf("Your Final Amount is: %.2f",final_price);
        }
        else{
            printf("No Discount\n");
            printf("Your Final Amount is: %2.f",cart_amount);
        }
    }
    return 0;
}
```

OutPut: -



```
E:\06Aug_Kevin_SE\C_Assign × + v
Enter the total cart amount:-3000
You got 20% Discount
Your Final Amount is: 2400.00

-----
Process exited after 3.774 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

Write a program that takes a user's age and checks eligibility for three things using if-else statements: if age ≥ 18 , print 'Eligible for Driving License'; if age ≥ 21 , also print 'Eligible for Credit Card'; if age ≥ 25 , also print 'Eligible for Car Rental'. Print all applicable messages for the given age.

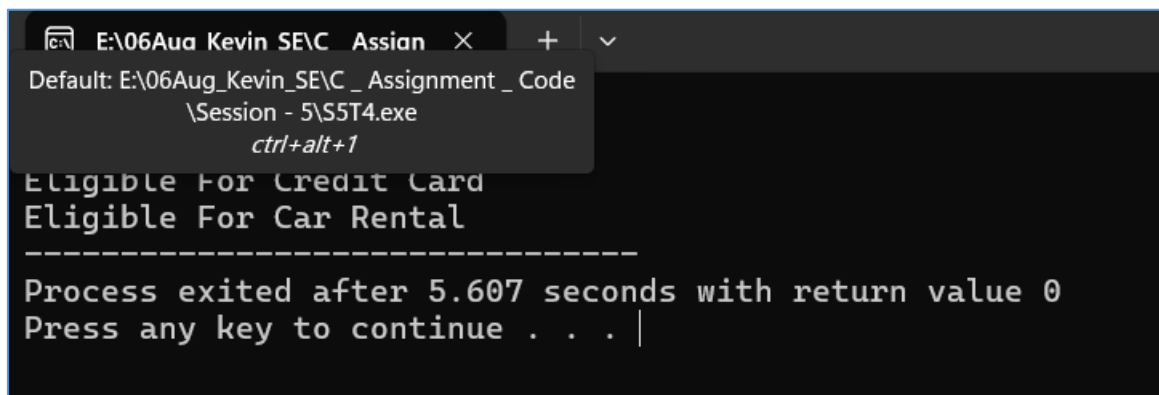
```
#include<stdio.h>
int main()
{
    int age;
    printf("Age:");
    scanf("%d",&age);

    if (age>=18)
    {
        printf("Lic\n");
    }

    if(age>=21)
    {
        printf("credit\n");
    }

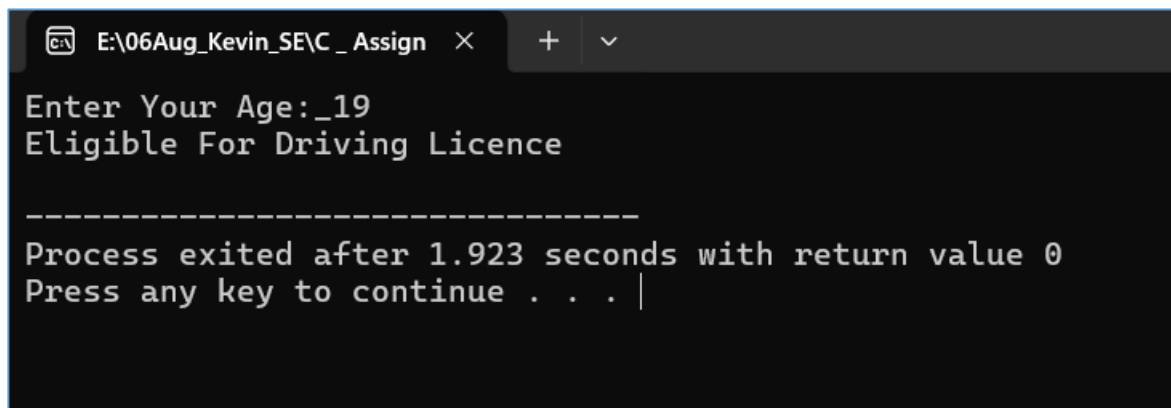
    if(age>=25)
    {
        printf("Car\n");
    }
}
else
{
    printf("not Applicable");
}
return 0 ;
}
```

Output: -



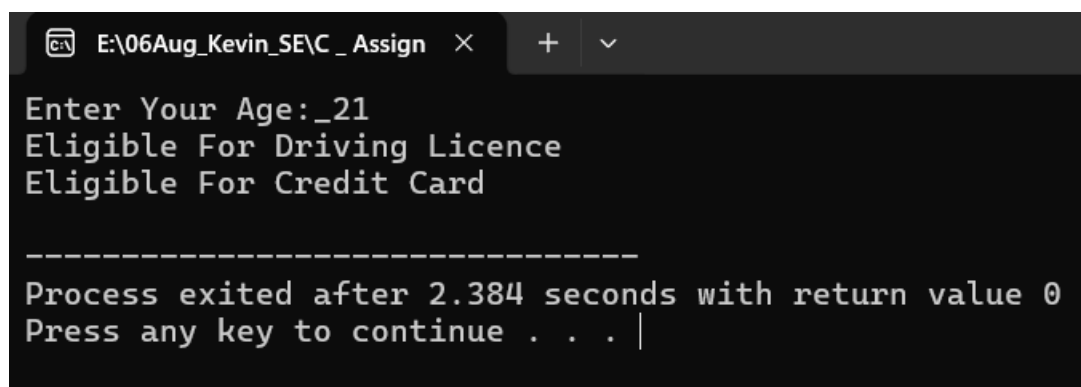
A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. A context menu is open, showing 'Default: E:\06Aug_Kevin_SE\C_Assignment_Code\S5T4.exe' and a keyboard shortcut 'ctrl+alt+1'. The command prompt displays the following text: 'Eligible For Credit Card', 'Eligible For Car Rental', a separator line '-----', 'Process exited after 5.607 seconds with return value 0', and 'Press any key to continue . . . |'.

```
E:\06Aug_Kevin_SE\C_Assign >
Default: E:\06Aug_Kevin_SE\C_Assignment_Code
\S5T4.exe
ctrl+alt+1
Eligible For Credit Card
Eligible For Car Rental
-----
Process exited after 5.607 seconds with return value 0
Press any key to continue . . . |
```



A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign'. The command prompt displays the following text: 'Enter Your Age:_19', 'Eligible For Driving Licence', a separator line '-----', 'Process exited after 1.923 seconds with return value 0', and 'Press any key to continue . . . |'.

```
E:\06Aug_Kevin_SE\C_Assign >
Enter Your Age:_19
Eligible For Driving Licence
-----
Process exited after 1.923 seconds with return value 0
Press any key to continue . . . |
```



A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign'. The command prompt displays the following text: 'Enter Your Age:_21', 'Eligible For Driving Licence', 'Eligible For Credit Card', a separator line '-----', 'Process exited after 2.384 seconds with return value 0', and 'Press any key to continue . . . |'.

```
E:\06Aug_Kevin_SE\C_Assign >
Enter Your Age:_21
Eligible For Driving Licence
Eligible For Credit Card
-----
Process exited after 2.384 seconds with return value 0
Press any key to continue . . . |
```

Session – 6

Que – 1

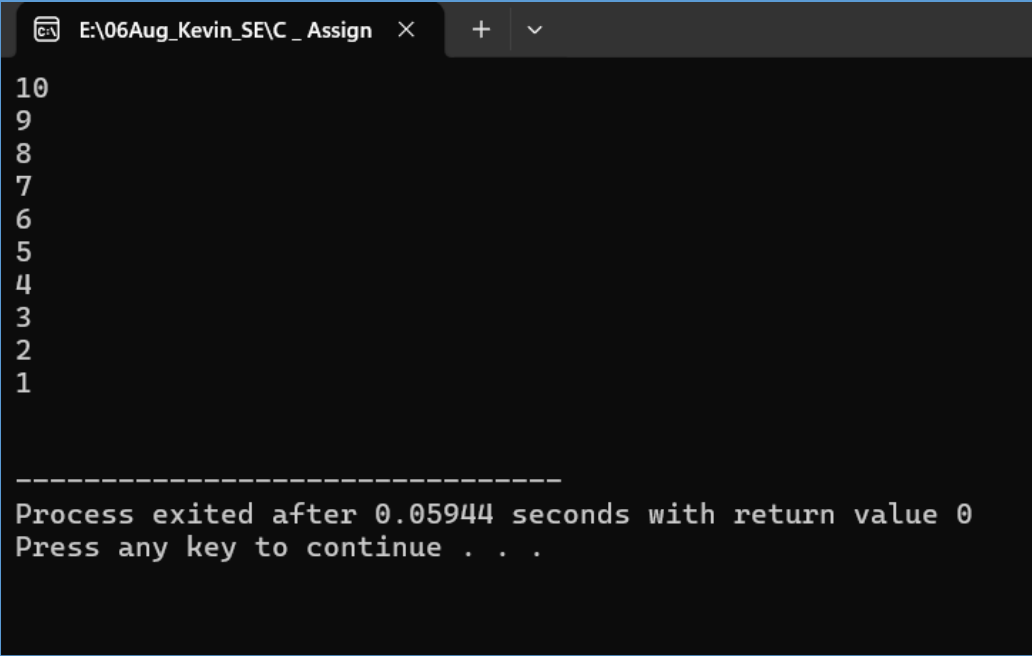
Write a program that prints the numbers from

10 down to 1 using a for loop,

similar to a countdown timer.

```
#include<stdio.h>
int main()
{
    int i;
    for (i=10;i>=1;i--)
    {
        printf("%d\n",i);
    }
    printf("\n");
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  X  +  v
10
9
8
7
6
5
4
3
2
1

-----
Process exited after 0.05944 seconds with return value 0
Press any key to continue . . .
```

Que – 2

Create a menu-driven console app that lets the user:

- 1) View your favorite 3 IPL teams,
- 2) Add a new team, 3) Exit.

Use a while loop to keep showing the menu until the user chooses Exit.

Hint: Use input() (or Scanner in Java) to get the user's choice each time.

```
#include<stdio.h>
int main()
{
    int choice;
    char team[15];
    while(1)
    {

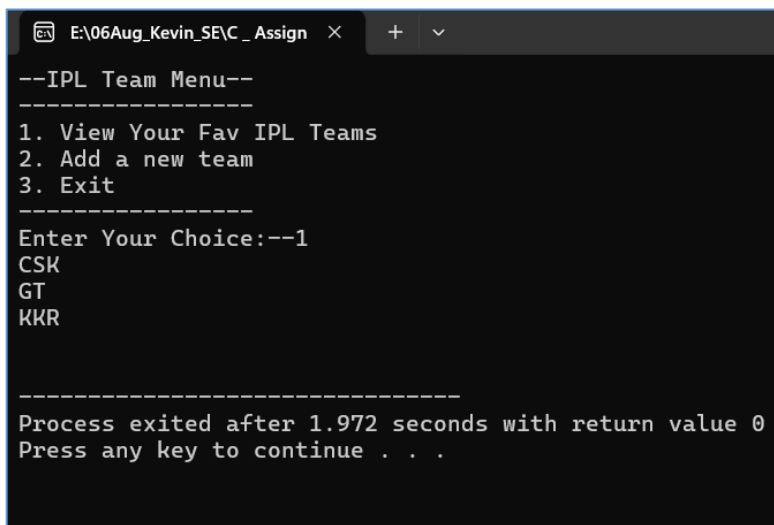
        printf("--IPL Team Menu--\n");
        printf("-----\n");
        printf("1. View Your Fav IPL Teams\n");
        printf("2. Add a new team\n");
        printf("3. Exit\n");
        printf("-----\n");
        printf("Enter Your CChoice:--");
        scanf("%d",&choice);

        switch (choice)
        {
            case 1:

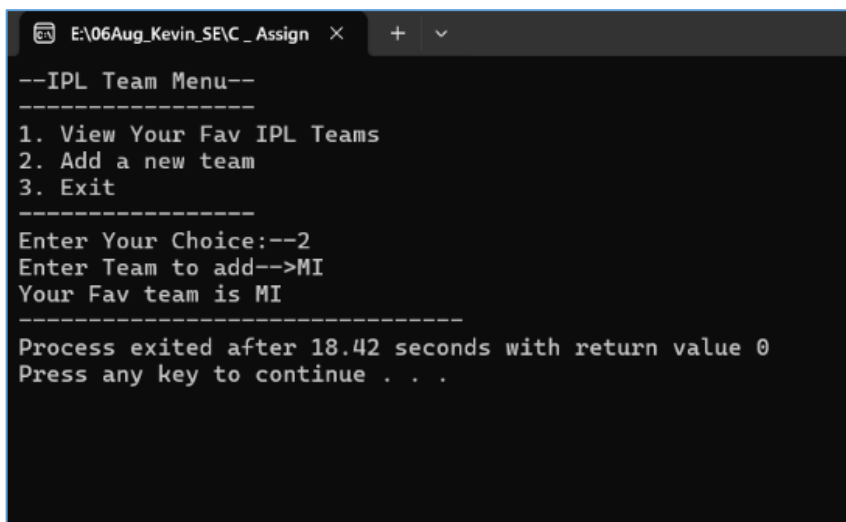
                // printf("Your Favorite IPL Teams are as below\n");
                printf("CSK\n");
                printf("GT\n");
                printf("KKR\n");
                printf("\n");
                break;
            case 2:
                printf("Enter Team to add-->");
                scanf("%s",team);
                printf("Your Fav team is %s",team);
                break;
            case 3:
```

```
        printf("Exited");  
        break;  
    }  
    return 0;  
}
```

Output →



```
E:\06Aug_Kevin_SE\C_Assign  X + v  
--IPL Team Menu--  
-----  
1. View Your Fav IPL Teams  
2. Add a new team  
3. Exit  
-----  
Enter Your Choice:--1  
CSK  
GT  
KKR  
  
-----  
Process exited after 1.972 seconds with return value 0  
Press any key to continue . . .
```



```
E:\06Aug_Kevin_SE\C_Assign  X + v  
--IPL Team Menu--  
-----  
1. View Your Fav IPL Teams  
2. Add a new team  
3. Exit  
-----  
Enter Your Choice:--2  
Enter Team to add-->MI  
Your Fav team is MI  
-----  
Process exited after 18.42 seconds with return value 0  
Press any key to continue . . .
```

Que – 3

Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it.

Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
#include<time.h>

int main ()
{
    char song1[50] = "Sunn";
    char song2[50] = "ShapeOfYou";
    char song3[50] = "Levitating";
    char guess[50];
    char random_pick[50];

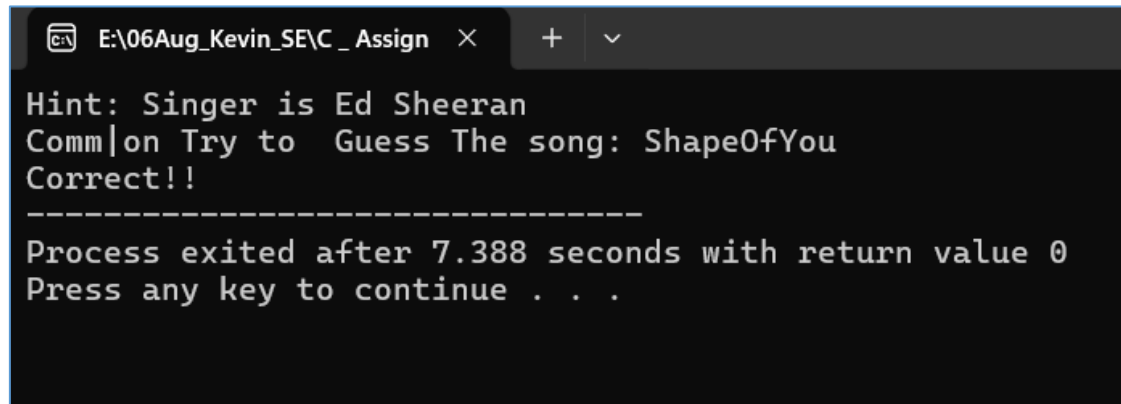
    int choice;
    srand(time(0));
    choice = rand()%3 ;

    switch(choice)
    {
        case 0:
            strcpy(random_pick,song1);
            printf("Hint: Singer is Dino James\n");
            break;
        case 1:
            strcpy(random_pick,song2);
            printf("Hint: Singer is Ed Sheeran\n");
            break;
        case 2:
            strcpy(random_pick,song3);
            printf("Hint: Singer is Dua Lipa\n");
            break;
    }

    do{
        printf("Comm|on Try to  Guess The song: ");
        scanf("%s",&guess);
```

```
        if (strcmp(guess,random_pick)==0)
        {
            printf("Correct!!");
            break;
        }
        else
        {
            printf("Wrong Guess Try Again\n");
        }
    }
    while(strcmp(guess,random_pick)!=0 );
    return 0;
}
```

Output –



```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ∨

Hint: Singer is Ed Sheeran
Comm|on Try to Guess The song: ShapeOfYou
Correct!!
-----
Process exited after 7.388 seconds with return value 0
Press any key to continue . . .
```

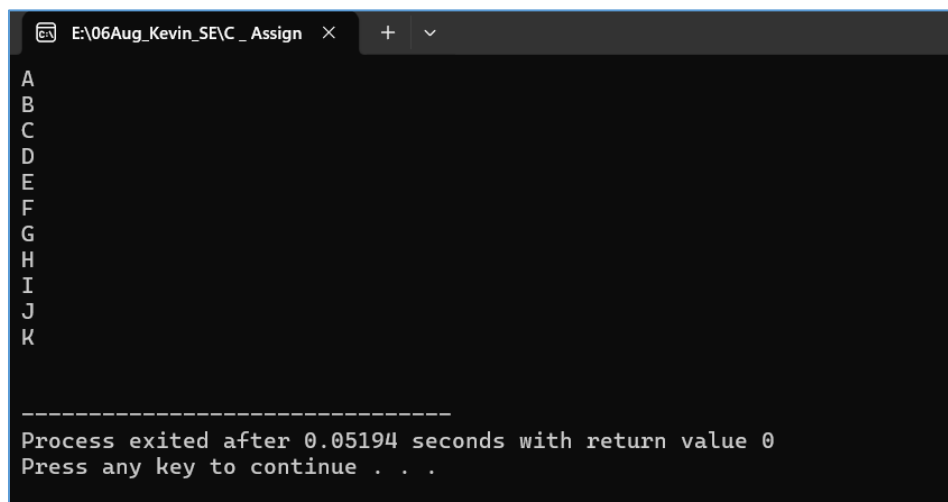
Que – 4

Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

```
// Entry Control Loops are: 1) while 2)for
// Exit Control Loops are: 1)for
An entry-controlled loop checks the condition before running the code
block,
while an exit-controlled loop runs the code block first before
checking the conditio
#include<stdio.h>
int main()
{
    char i='A';

    while (i<='Z')
    {
        printf("%c\n",i);
        i++;
    }
    printf("\n");

    return 0;
}
```

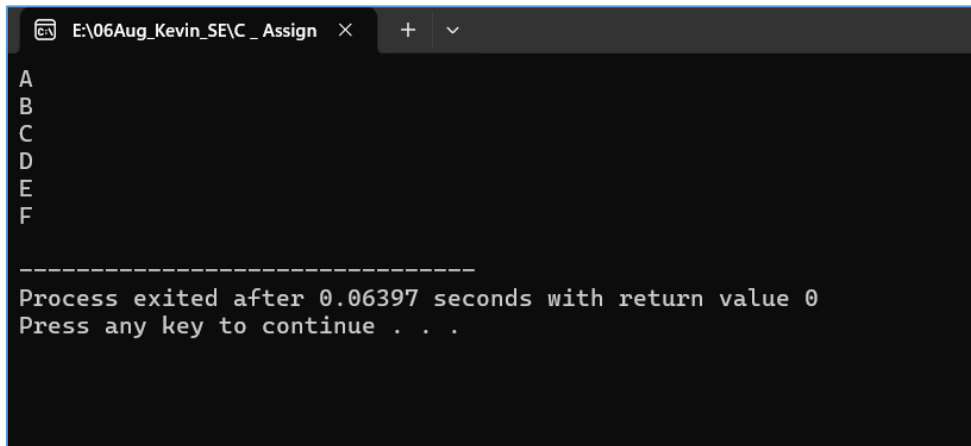
A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign'. The window contains the output of a C program. It displays the characters 'A' through 'K' on separate lines. Below the characters, there is a separator line followed by the text 'Process exited after 0.05194 seconds with return value 0' and 'Press any key to continue . . .'.

```
E:\06Aug_Kevin_SE\C_Assign >
A
B
C
D
E
F
G
H
I
J
K

-----
Process exited after 0.05194 seconds with return value 0
Press any key to continue . . .
```

// Do - While Exit Control loop

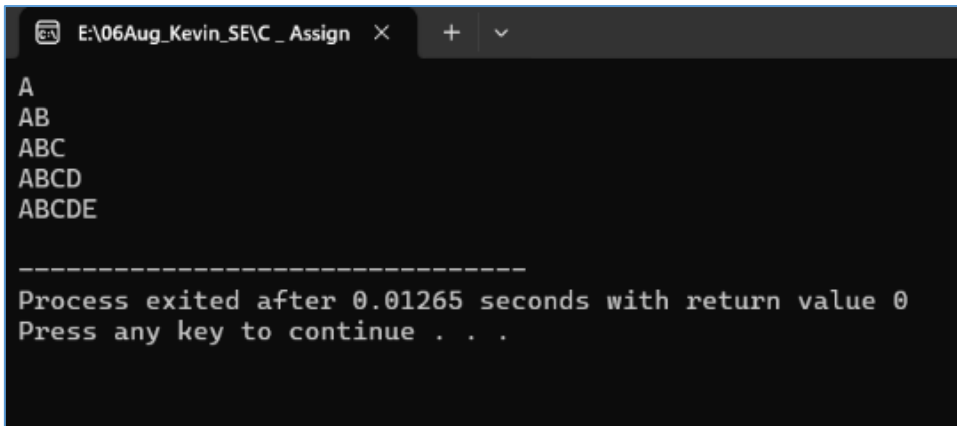
```
#include<stdio.h>
int main()
{
    char i = 'A';
    do {
        printf("%c\n",i);
        i++;
    }
    while (i<='F');
}
```



```
E:\06Aug_Kevin_SE\C_Assign
A
B
C
D
E
F
-----
Process exited after 0.06397 seconds with return value 0
Press any key to continue . . .
```

```
// for Entry control loop

#include<stdio.h>
int main()
{
    int i,j;
    char ch;
    for(i=1;i<=5;i++)
    {
        ch = 'A';
        for(j=1;j<=i;j++)
        {
            printf("%c",ch);
            ch++;
        }
        printf("\n");
    }
}
```



The screenshot shows a terminal window with a dark background. The title bar at the top reads "E:\06Aug_Kevin_SE\C_Assign" followed by a close button and window control buttons. The output of the program is displayed as follows:

```
A
AB
ABC
ABCD
ABCDE

-----
Process exited after 0.01265 seconds with return value 0
Press any key to continue . . .
```

Session – 7

Que – 1

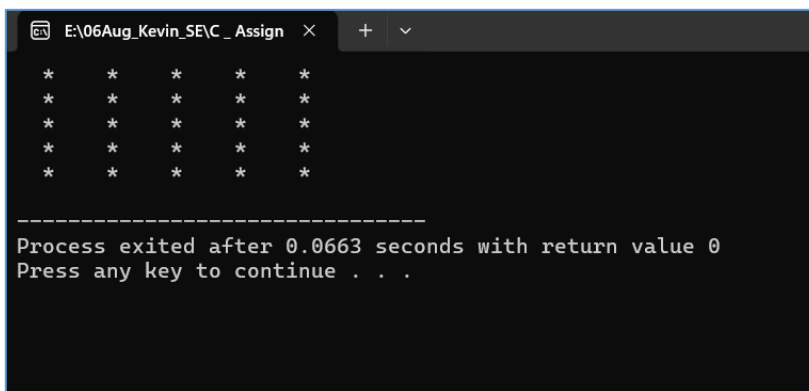
Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 🍷 symbol.

```
#include <stdio.h>

int main()
{
    for(int i=1;i<=5;i++)
    {
        for(int j =1;j<=5;j++)
        {
            printf("  *  ");
        }

        printf("\n");
    }
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign >
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *

-----
Process exited after 0.0663 seconds with return value 0
Press any key to continue . . .
```

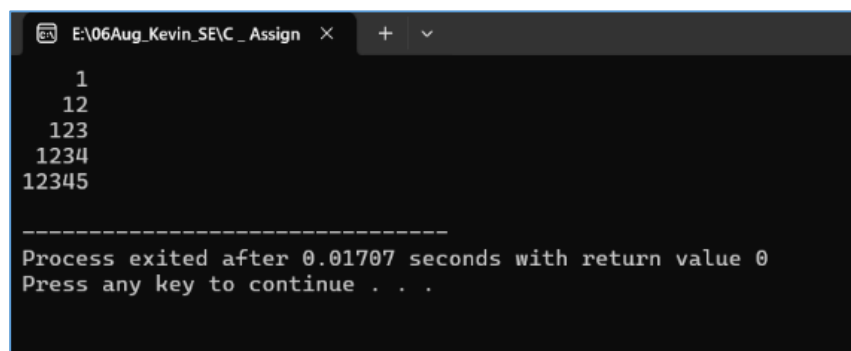
Que – 2

Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

```
#include<stdio.h>
int main()
{
    int i,j,k;
    int n=5;

    for (i=1;i<=n;i++)
    {
        for(j=n-1;j>=i;j--)
        {
            printf(" ");
        }
        for(k=1;k<=i;k++)
        {
            printf("%d",k);
        }
        printf("\n");
    }
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C _ Assign  >
1
12
123
1234
12345

-----
Process exited after 0.01707 seconds with return value 0
Press any key to continue . . .
```

Que – 3

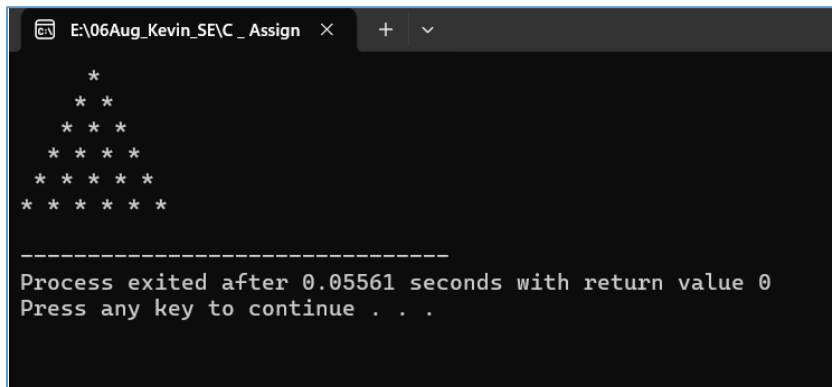
Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.

Hint: Use spaces to align the stars in the center for each row.

```
#include<stdio.h>
int main()
{
    int i,j,k;
    int n=6;

    for (i=1;i<=n;i++)
    {
        for(j=n-1;j>=i;j--)
        {
            printf(" ");
        }
        for(k=1;k<=i;k++)
        {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  X + v
  *
 * *
* * *
* * * *
* * * * *
* * * * * *

-----
Process exited after 0.05561 seconds with return value 0
Press any key to continue . . .
```

Que – 4

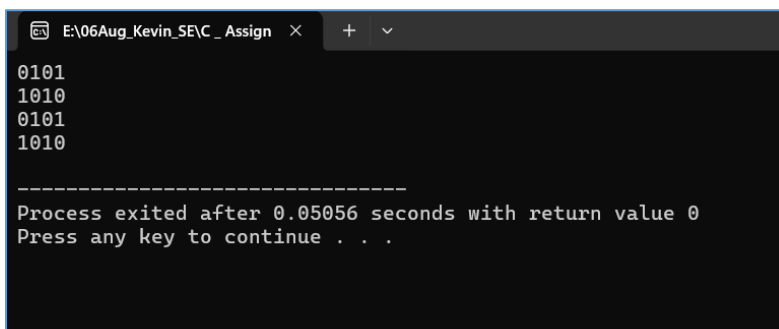
Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```
#include <stdio.h>

int main()
{
    for(int i=1;i<=4;i++)
    {
        for(int j =1;j<=4;j++)
        {
            if ((i+j)%2 == 0)
            {
                printf("0");
            }
            else
            {
                printf("1");
            }
        }

        printf("\n");
    }
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign >
0101
1010
0101
1010

-----
Process exited after 0.05056 seconds with return value 0
Press any key to continue . . .
```

Que – 5

Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

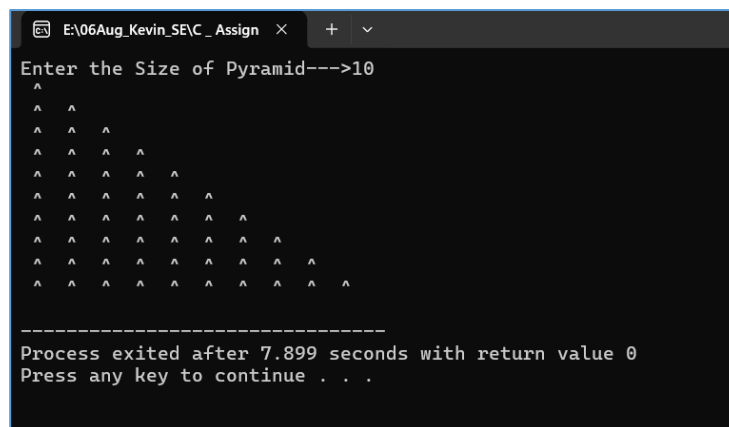
```
#include <stdio.h>

int main()
{
    int size ;
    printf("Enter the Size of Pyramid--->");
    scanf("%d",&size);

    for(int i=1;i<=size;i++)
    {
        for(int j=1;j<=i;j++)
        {
            printf(" ^ ");
        }

        printf("\n");
    }
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign >
Enter the Size of Pyramid--->10
^
^ ^
^ ^ ^
^ ^ ^ ^
^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^

-----
Process exited after 7.899 seconds with return value 0
Press any key to continue . . .
```

Session 8

Que – 1

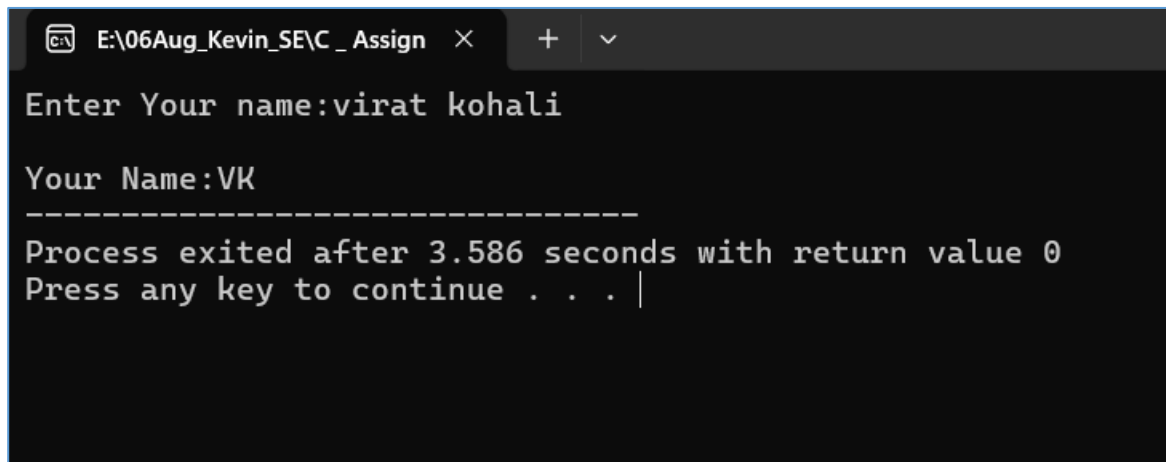
```
/*  
    Declare a function called getUserInitials  
    that takes a user's full name (like 'Virat Kohli')  
    and returns their initials in uppercase (e.g., 'VK').  
    Call this function with your favorite cricketer's name and print  
    the result.  
*/
```

```
#include<stdio.h>  
#include<string.h>  
#include<ctype.h>
```

```
void getUSeInit()  
{  
    char name[100];  
    int i;  
    printf("Enter Your name:");  
    gets(name);  
    printf("\nYour Name:%c",toupper(name[0]));  
  
    for(i=0;name[i] != '\0';i++)  
    {  
        if(name[i] == ' ')  
        {  
            printf("%c",toupper(name[i+1]));  
        }  
    }  
}
```

```
int main()  
{  
    getUSeInit();  
    return 0;  
}
```

Output :



```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ▾  
Enter Your name:virat kohali  
  
Your Name:VK  
-----  
Process exited after 3.586 seconds with return value 0  
Press any key to continue . . . |
```

Que – 2

```
/*  
    Create a function addToCart that takes  
    a shopping cart array and a product name,  
    adds the product to the cart, and prints the updated cart.  
    Demonstrate how passing the cart  
    array by reference allows changes  
    to persist outside the function.
```

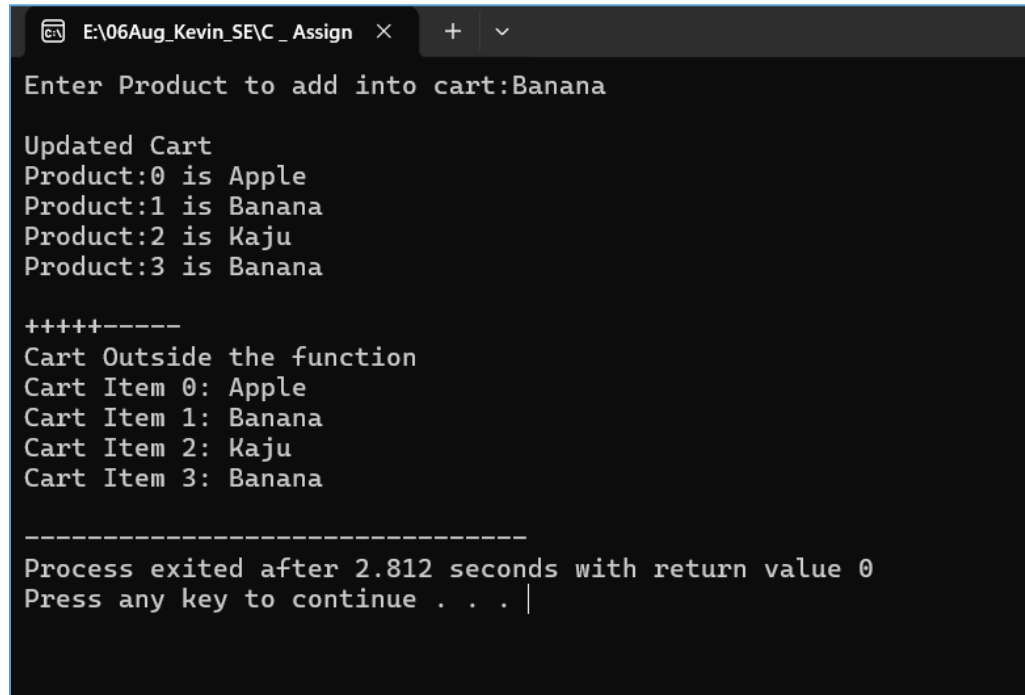
```
    Hint: In languages like JavaScript,  
    arrays are passed by reference.  
    In C/C++, use pointers for reference behavior.
```

```
*/
```

```
#include <stdio.h>  
#include <string.h>  
  
void addToCart(char (*cart)[50],char product[10],int *count)  
{  
    strcpy(cart[*count],product);  
    (*count)++;  
  
    printf("\nUpdated Cart\n");  
    for(int i = 0; i<*count;i++)  
    {  
        printf("Product:%d is %s\n",i,card[i]);  
    }  
}  
  
int main()  
{  
    char cart[10][50] = {"Apple","Banana","Kaju"};  
    char product[10];  
    int count = 3;  
    printf("Enter Product to add into cart:");  
    scanf("%s",product);  
    addToCart(cart,product,&count);  
    printf("\n+++++----\n");  
    printf("Cart Outside the function\n");  
    for(int i=0;i<count;i++)  
    {  
        printf("Cart Item %d: %s\n",i,card[i]);  
    }  
}
```

```
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign > Enter Product to add into cart:Banana

Updated Cart
Product:0 is Apple
Product:1 is Banana
Product:2 is Kaju
Product:3 is Banana

+++++-----
Cart Outside the function
Cart Item 0: Apple
Cart Item 1: Banana
Cart Item 2: Kaju
Cart Item 3: Banana

-----
Process exited after 2.812 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

```
/*
    Write two functions:
    increaseFollowersByValue and increaseFollowersByReference.
    Each should take a followers count and add 1000 to it,
    but one should use pass-by-value and the other pass-by-reference.
    Show how the original followers count changes (or doesn't) after
    calling each function.
*/

#include <stdio.h>

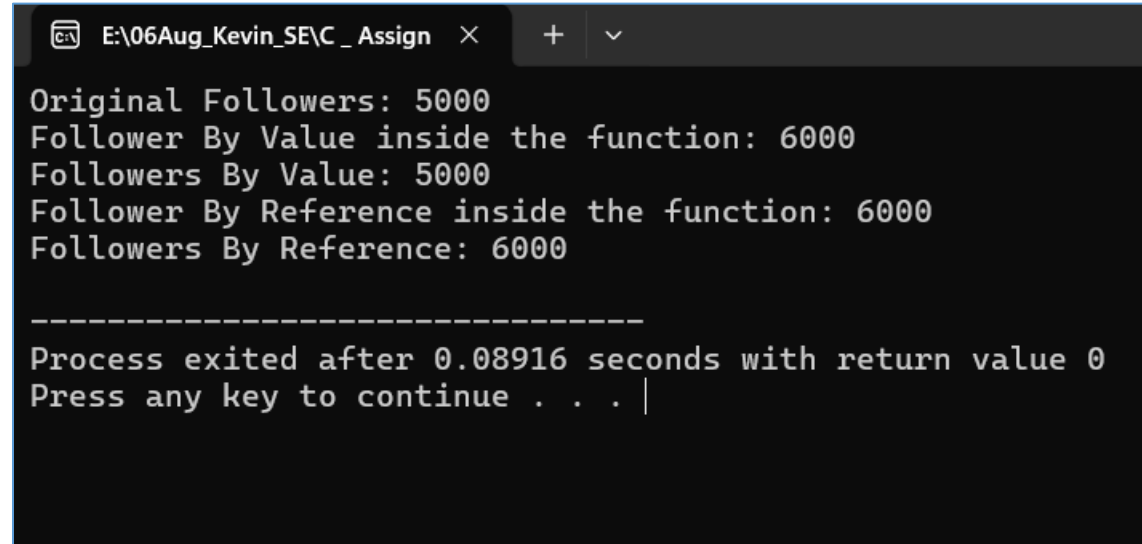
int followByValue(int follower)
{
    //follower = 1000;
    follower = follower + 1000;
    printf("Follower By Value inside the function: %d\n",follower);
}

int followByRef(int *follower)
{
    /**follower = 1000;
    *follower = *follower + 1000;
    printf("Follower By Reference inside the function:
%d\n",*follower);
}
int main()
{

    int follower = 5000;
    printf("Original Followers: %d\n",follower);
    followByValue(follower);
    printf("Followers By Value: %d\n",follower);
    followByRef(&follower);
    printf("Followers By Reference: %d\n",follower);

    return 0;
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. The output text is as follows:

```
Original Followers: 5000
Follower By Value inside the function: 6000
Followers By Value: 5000
Follower By Reference inside the function: 6000
Followers By Reference: 6000

-----
Process exited after 0.08916 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

```
/*
    Build a function formatPrice that takes a price
    in rupees and returns a string formatted like Flipkart's price tag
    (e.g., '$1,599').
    Use this function to display prices for three different products.
*/

#include <stdio.h>

void formatPrice(int price, char result[])
{
    if (price >= 1000)
    {
        sprintf(result, "%d,%03d", price / 1000, price % 1000);
    }
    else
    {
        sprintf(result, "%d", price);
    }
}

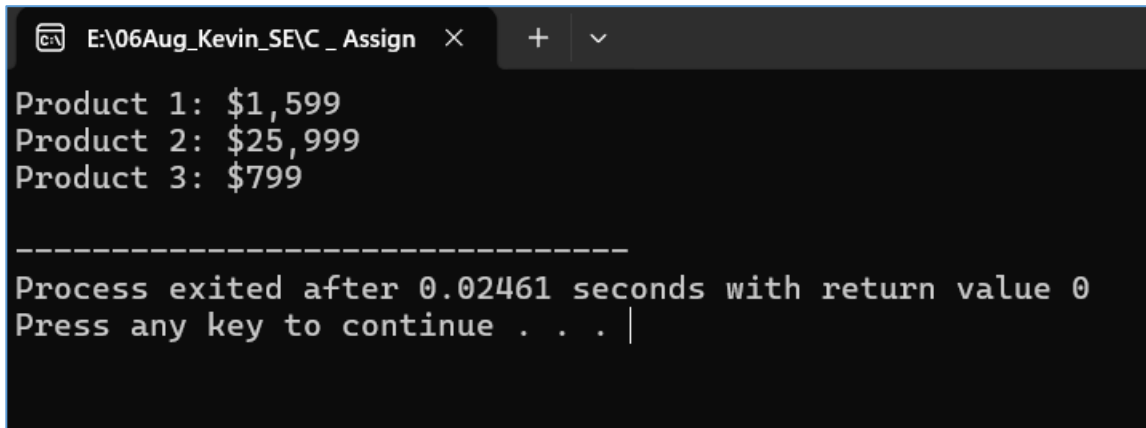
int main()
{
    char price1[20];
    char price2[20];
    char price3[20];

    formatPrice(1599, price1);
    formatPrice(25999, price2);
    formatPrice(799, price3);

    printf("Product 1: $%s\n", price1);
    printf("Product 2: $%s\n", price2);
    printf("Product 3: $%s\n", price3);

    return 0;
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. The command prompt displays the following text: 'Product 1: \$1,599', 'Product 2: \$25,999', and 'Product 3: \$799'. Below these, a line of dashes separates the output from the exit message: 'Process exited after 0.02461 seconds with return value 0'. The final line is 'Press any key to continue . . . |', with a vertical cursor at the end.

```
E:\06Aug_Kevin_SE\C_Assign >
Product 1: $1,599
Product 2: $25,999
Product 3: $799
-----
Process exited after 0.02461 seconds with return value 0
Press any key to continue . . . |
```

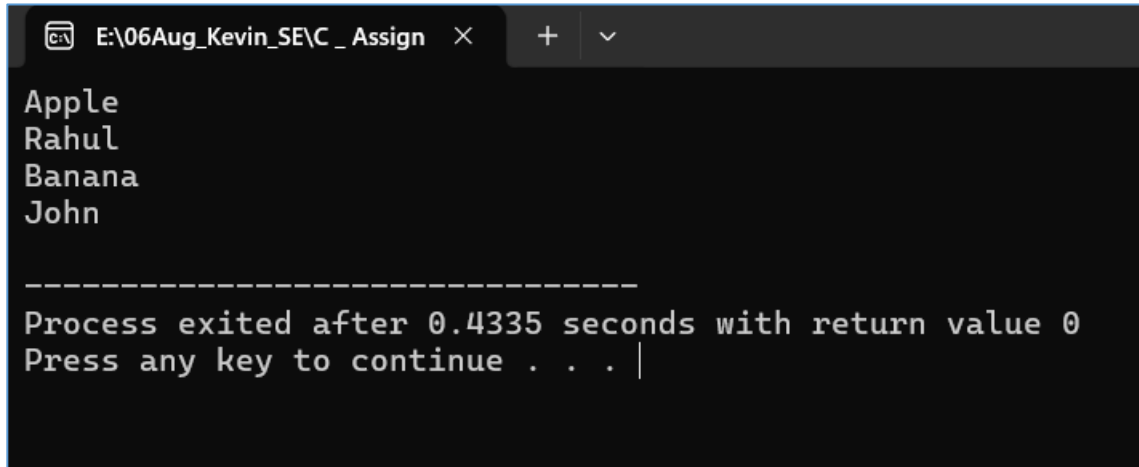
Que – 5

```
/*
    Refactor an existing function you wrote above to
    make it reusable for both product names and usernames
    (for example, a function that capitalizes the first letter of any
    string).
    Constraint:
    The refactored function should work for any string input, not just
    a specific use case.
*/
#include <stdio.h>
#include <ctype.h>

void capitalize(char str[])
{
    str[0] = toupper(str[0]);
}

int main()
{
    char names[4][50] = {
        "apple",
        "rahul",
        "banana",
        "john"
    };
    for (int i = 0; i < 4; i++)
    {
        capitalize(names[i]);
        printf("%s\n", names[i]);
    }
    return 0;
}
```

Output:



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'E:\06Aug_Kevin_SE\C_Assign' and includes standard window controls. The output of the program is displayed in white text, listing 'Apple', 'Rahul', 'Banana', and 'John' on separate lines. A horizontal dashed line separates the list from the final status message: 'Process exited after 0.4335 seconds with return value 0'. The prompt concludes with 'Press any key to continue . . . |', where the vertical bar represents the cursor.

```
E:\06Aug_Kevin_SE\C_Assign > Apple
Rahul
Banana
John

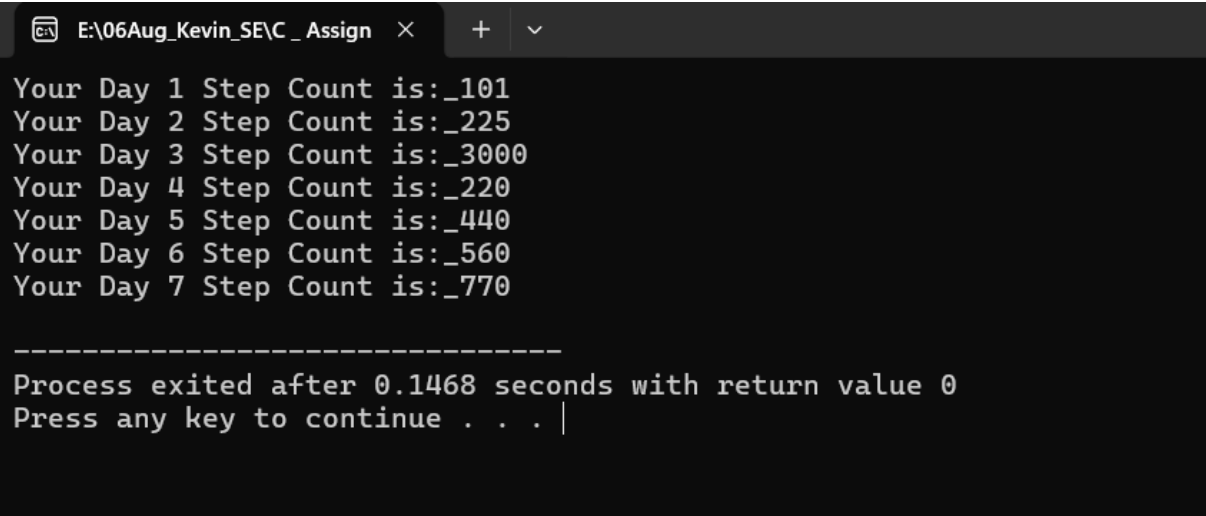
-----
Process exited after 0.4335 seconds with return value 0
Press any key to continue . . . |
```

Session 9

Que – 1

```
/*  
    Declare a 1D array called dailySteps with 7  
    elements to store your step count  
    for each day of the week,  
    assign sample values,  
    and print each value using a loop.  
*/  
#include<stdio.h>  
int main()  
{  
    int daily_steps[8] = {0,101,225,3000,220,440,560,770};  
    int i;  
  
    for(i=1;i<=7;i++)  
    {  
        printf("Your Day %d Step Count is:_%d\n",i,daily_steps[i]);  
    }  
}
```

OutPut:



```
E:\06Aug_Kevin_SE\C_Assign  X  +  v  
Your Day 1 Step Count is:_101  
Your Day 2 Step Count is:_225  
Your Day 3 Step Count is:_3000  
Your Day 4 Step Count is:_220  
Your Day 5 Step Count is:_440  
Your Day 6 Step Count is:_560  
Your Day 7 Step Count is:_770  
  
-----  
Process exited after 0.1468 seconds with return value 0  
Press any key to continue . . . |
```

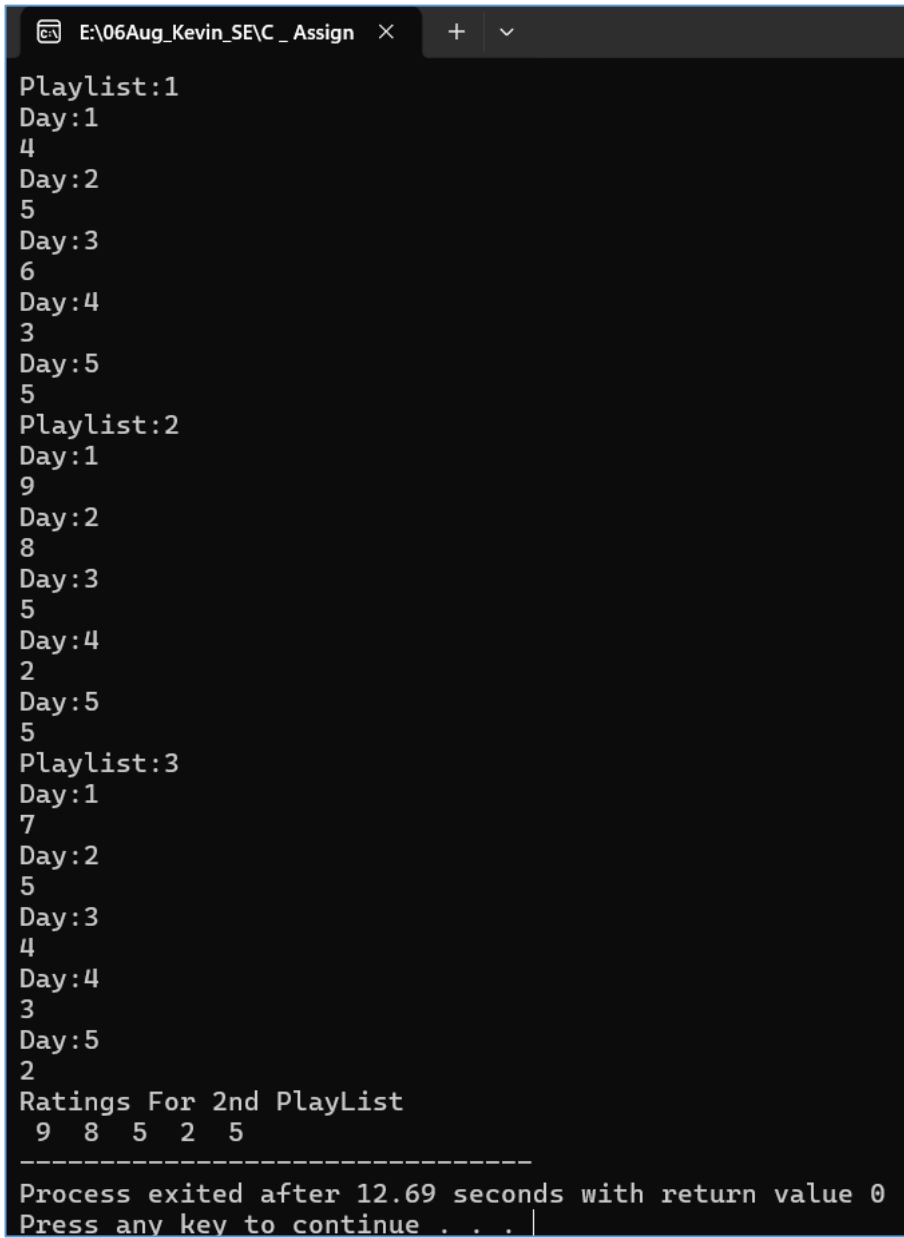
Que – 2

```
/*
    Create a 2D array called
    playlistRatings to store ratings for 3 Spotify playlists
    over 5 days (rows = playlists, columns = days).
    Fill it with sample numbers
    and print the ratings for the second playlist.
*/

#include<stdio.h>
int main()
{
    int playListRatings[3][5];
    int i,j;

    for (i=0;i<3;i++)
    {
        printf("Playlist:%d\n",i+1);
        for(j=0;j<=4;j++)
        {
            printf("Day:%d\n",j+1);
            scanf("%d",&playListRatings[i][j]);
        }
    }
    printf("Ratings For 2nd PlayList\n");
    for(j=0;j<5;j++)
    {
        printf(" %d ",playListRatings[1][j]);
    }
}
```

Output:

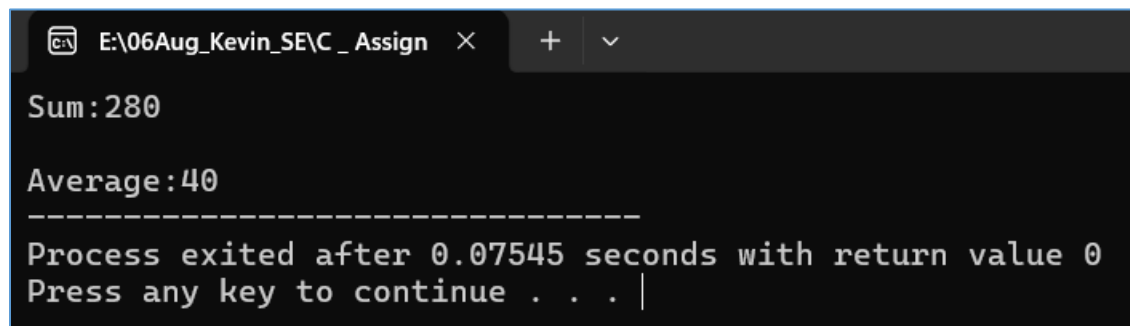
A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C _ Assign' and standard window controls. The output text is as follows:

```
Playlist:1
Day:1
4
Day:2
5
Day:3
6
Day:4
3
Day:5
5
Playlist:2
Day:1
9
Day:2
8
Day:3
5
Day:4
2
Day:5
5
Playlist:3
Day:1
7
Day:2
5
Day:3
4
Day:4
3
Day:5
2
Ratings For 2nd PlayList
9 8 5 2 5
-----
Process exited after 12.69 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

```
/*
    Build a function that takes a 1D array of 7 integers
    representing your daily Zomato order amounts and calculates
    the average spend for the week.
    Hint:
    Use a loop to sum the values, then divide by the array length.
*/
#include<stdio.h>
seven()
{
    int days[7] = {10,20,30,40,50,60,70};
    float sum = 0;
    float avg_spend ;
    int i;
    for(i=0;i<7;i++)
    {
        sum = sum + days[i];
    }
    printf("Sum:%.0f\n\n",sum);
    avg_spend = sum/7;
    printf("Average:%.0f",avg_spend);
}
int main()
{
    seven();
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C _ Assign  X  +  v
Sum: 280
Average: 40
-----
Process exited after 0.07545 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

```
/*
    Given a 2D array called cricketScores
    where each row represents an IPL
    match and columns represent runs scored
    by each team,
    write code to print the highest score from each match.
*/

#include<stdio.h>
int main()
{
    int cricketScores[3][3];
    int i,j,high;

    for (i=0;i<3;i++)
    {
        printf("Number of Match:%d\n",i+1);
        for(j=0;j<3;j++)
        {
            printf("Enter Score of Team %d:",j+1);
            scanf("%d",&cricketScores[i][j]);
        }
    }

    printf("\tCricket Scores\n");

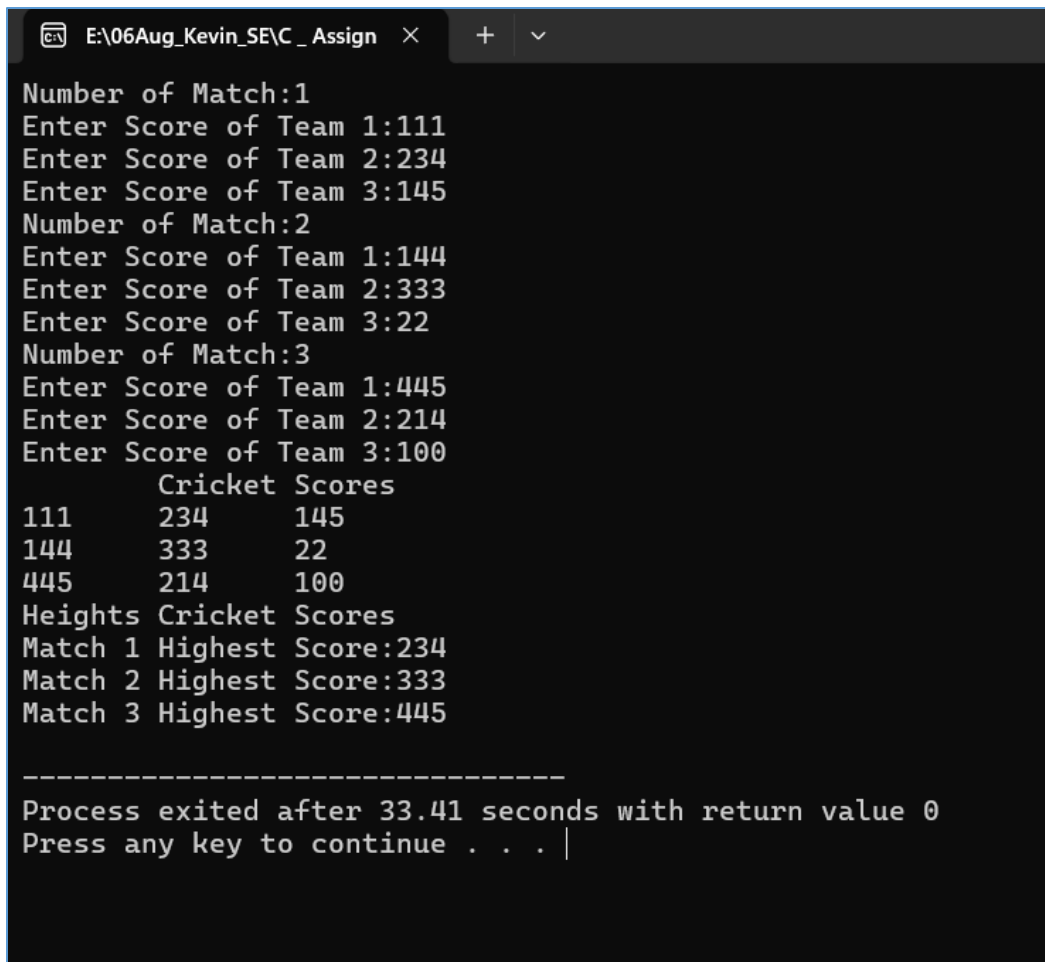
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t",cricketScores[i][j]);
        }
        printf("\n");
    }

    printf("Heights Cricket Scores\n");

    for(i=0;i<3;i++)
    {
        high = cricketScores[i][0];
        for(j=0;j<3;j++)
        {
            if(cricketScores[i][j]>high)
```

```
        {  
            high = cricketScores[i][j];  
        }  
    }  
    printf("Match %d Highest Score:%d\n",i+1,high);  
}  
return 0;  
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign >
Number of Match:1
Enter Score of Team 1:111
Enter Score of Team 2:234
Enter Score of Team 3:145
Number of Match:2
Enter Score of Team 1:144
Enter Score of Team 2:333
Enter Score of Team 3:22
Number of Match:3
Enter Score of Team 1:445
Enter Score of Team 2:214
Enter Score of Team 3:100
      Cricket Scores
111      234      145
144      333      22
445      214      100
Heights Cricket Scores
Match 1 Highest Score:234
Match 2 Highest Score:333
Match 3 Highest Score:445

-----
Process exited after 33.41 seconds with return value 0
Press any key to continue . . . |
```

Session - 10

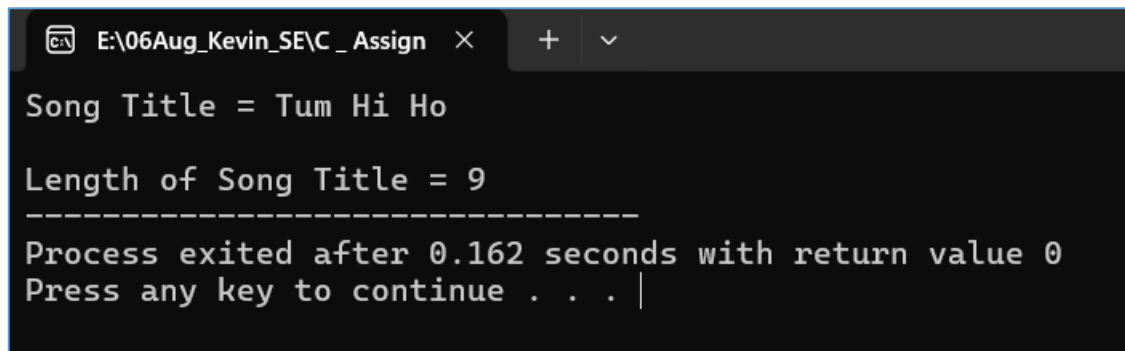
Que – 1

```
/*
    Declare a string variable called
    songTitle and assign it the value 'Tum Hi Ho'.
    Print the length of the string using strlen().
*/

#include<stdio.h>
#include<string.h>

int main()
{
    char songTitle[] = "Tum Hi Ho";
    printf("Song Title = %s",songTitle);
    printf("\n");
    printf("\nLength of Song Title = %d",strlen(songTitle));
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  X  +  v
Song Title = Tum Hi Ho
Length of Song Title = 9
-----
Process exited after 0.162 seconds with return value 0
Press any key to continue . . . |
```

Que – 2

```
/*
    Take input for two usernames (as strings)
    and compare them using strcmp().
    Display whether they are the same or different.
*/

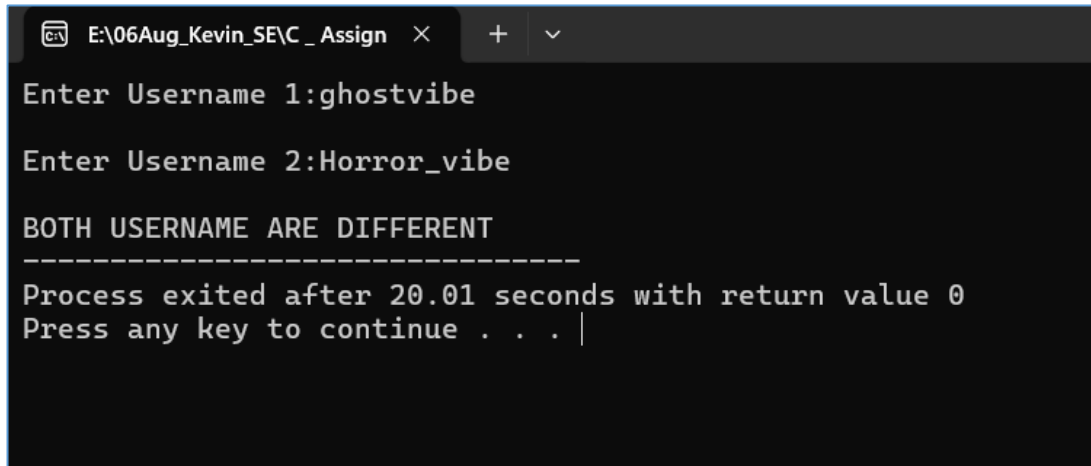
#include<stdio.h>
#include<string.h>

int main()
{
    char username_one[10];
    char username_two[10];

    printf("Enter Username 1:");
    scanf("%s",username_one);
    printf("\nEnter Username 2:");
    scanf("%s",username_two);

    if(strcmp(username_one,username_two)==0)
    {
        printf("\nBOTH USERNAME ARE SAME");
    }
    else
    {
        printf("\nBOTH USERNAME ARE DIFFERENT");
    }
    return 0;
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. The command prompt displays the following text: 'Enter Username 1:ghostvibe', 'Enter Username 2:Horror_vibe', 'BOTH USERNAME ARE DIFFERENT', a line of dashes '-----', 'Process exited after 20.01 seconds with return value 0', and 'Press any key to continue . . . |' with a cursor at the end.

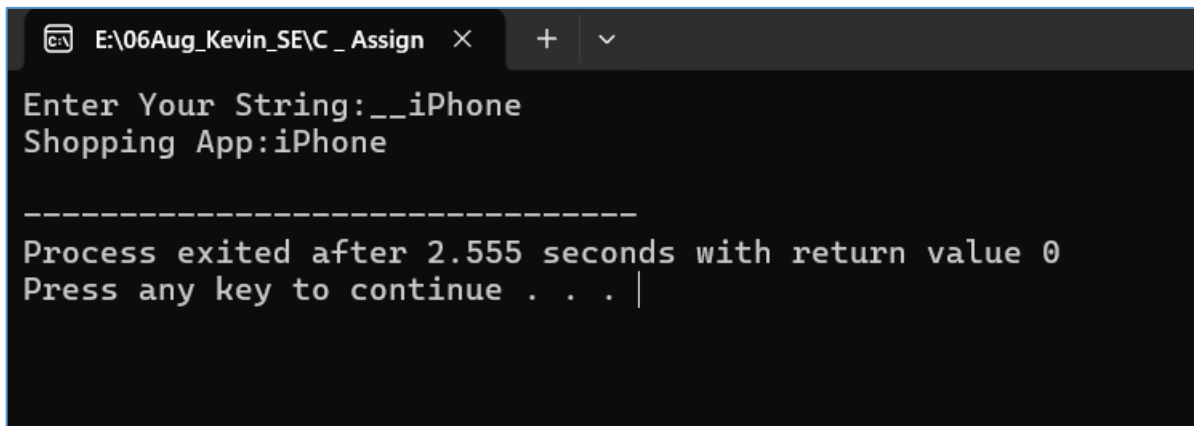
```
E:\06Aug_Kevin_SE\C_Assign > Enter Username 1:ghostvibe
Enter Username 2:Horror_vibe
BOTH USERNAME ARE DIFFERENT
-----
Process exited after 20.01 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

```
/*  
    Copy the string 'Flipkart' into another string  
    Make sure to declare enough space for the destination string.  
    variable called shoppingApp using strcpy(),  
    then print the value of shoppingApp.  
    Hint:  
    Make sure to declare enough space for the destination string.  
*/
```

```
#include<stdio.h>  
#include<string.h>  
int main()  
{  
    char str1[10];  
    char Shopping_app[10];  
    printf("Enter Your String:__");  
    scanf("%s",str1);  
    printf("Your Entered String: %s\n",str1);  
    printf("Shopping App:%s\n",strcpy(Shopping_app,str1));  
    return 0;  
}
```

Output:



```
E:\06Aug_Kevin_SE\C _ Assign  ×  +  ▾  
  
Enter Your String: __iPhone  
Shopping App:iPhone  
  
-----  
Process exited after 2.555 seconds with return value 0  
Press any key to continue . . . |
```

Que – 4

```
/*
    Build a small script that takes the
    user's full name as input
    and creates a username by copying only
    the first 5 characters using strcpy().
    Print the generated username.
    Constraint:
    If the name is shorter than 5 characters,
    use the full name as the username.
*/
#include<stdio.h>
#include<string.h>

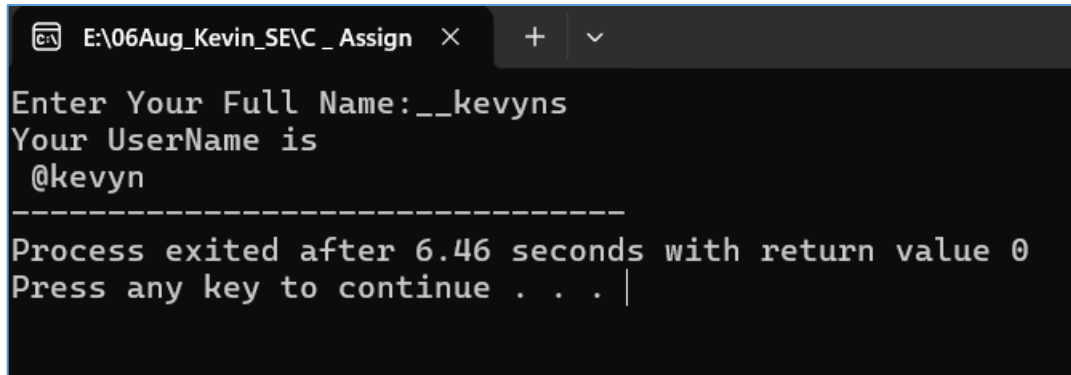
int main()
{
    char full_name[10];
    char username[10];

    int len;
    printf("Enter Your Full Name:__");
    gets(full_name);

    len = strlen(full_name);

    if (len >=5)
    {
        full_name[5] = '\0';
        printf("Your UserName is \n @%s",strcpy(username,full_name));
    }
    else
    {
        printf("Your Username is \n @%s",full_name);
    }
    return 0;
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. The command prompt displays the following text: 'Enter Your Full Name: __kevyns', 'Your UserName is', '@kevyn', a line of dashes '-----', 'Process exited after 6.46 seconds with return value 0', and 'Press any key to continue . . . |' with a cursor at the end.

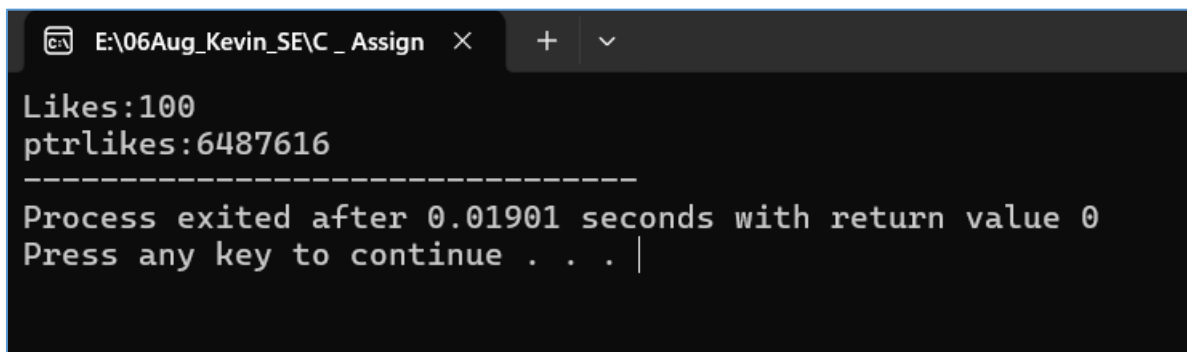
```
E:\06Aug_Kevin_SE\C_Assign > Enter Your Full Name: __kevyns
Your UserName is
@kevyn
-----
Process exited after 6.46 seconds with return value 0
Press any key to continue . . . |
```

Session - 11

Que – 1

```
/*  
    Declare an integer variable called likes and a pointer  
    variable called ptrLikes;  
    assign likes a value,  
    point ptrLikes to likes,  
    and print both the value and the address stored in ptrLikes.  
*/  
  
#include<stdio.h>  
int main()  
{  
    int like;  
    like = 100;  
    int *ptrlikes = &like;  
    printf("Likes:%d\n",like);  
    printf("ptrlikes:%d",ptrlikes);  
}
```

Output:



```
E:\06Aug_Kevin_SE\C _ Assign  X  +  v  
  
Likes:100  
ptrlikes:6487616  
-----  
Process exited after 0.01901 seconds with return value 0  
Press any key to continue . . . |
```

Que – 2

```
/*
    Write a function swapPlaylistCounts(int *a, int *b)
    that swaps the number of songs in two Spotify playlists
    using pointers, then call the function in main and print the
    swapped values.
*/

#include <stdio.h>

void swap(int *a, int *b)
{
    int temp;

    temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int A = 25;
    int B = 40;

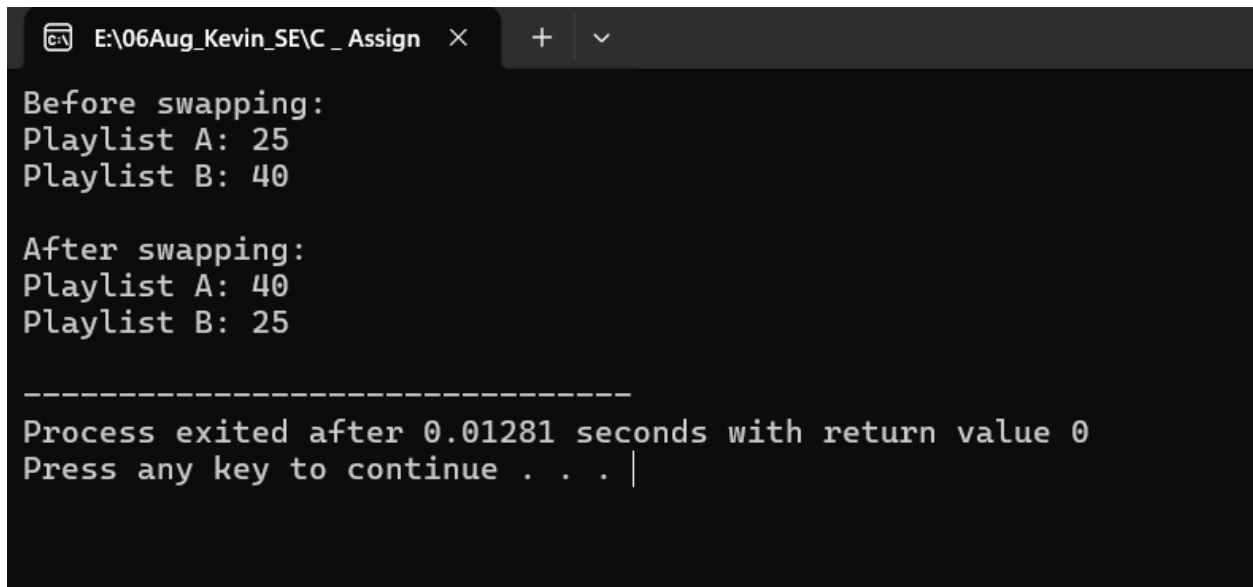
    printf("Before swapping:\n");
    printf("Playlist A: %d\n", A);
    printf("Playlist B: %d\n", B);

    swap(&A, &B);

    printf("\nAfter swapping:\n");
    printf("Playlist A: %d\n", A);
    printf("Playlist B: %d\n", B);

    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign × + v
Before swapping:
Playlist A: 25
Playlist B: 40

After swapping:
Playlist A: 40
Playlist B: 25

-----
Process exited after 0.01281 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

```
/*  
    Given an array of 5 order amounts (e.g., Zomato orders),  
    use a pointer to iterate through the array  
    and print each amount along with its memory address.  
    Hint: Use pointer arithmetic to move to the next element.  
*/  
  
#include<stdio.h>  
int main()  
{  
    int order[5],i;  
  
    for(i=1;i<=5;i++)  
    {  
        printf("ENTER_ORDER_AMMOUNT_%d:",i);  
        scanf("%d",&order[i]);  
  
    }  
    for(i=1;i<=5;i++)  
    {  
        printf("OrderAmount_%d : %d\n\n",i,order[i]);  
        printf("Memory_Address of OrderAmount_%d :  
%d\n\n",i,&order[i]);  
    }  
    return 0;  
}
```

Output:

```
E:\06Aug_Kevin_SE\C_Assign  X  +  v
ENTER_ORDER_AMMOUNT_1:100
ENTER_ORDER_AMMOUNT_2:300
ENTER_ORDER_AMMOUNT_3:400
ENTER_ORDER_AMMOUNT_4:200
ENTER_ORDER_AMMOUNT_5:421
OrderAmount_1 : 100

Memory_Address of OrderAmount_1 : 6487604

OrderAmount_2 : 300

Memory_Address of OrderAmount_2 : 6487608

OrderAmount_3 : 400

Memory_Address of OrderAmount_3 : 6487612

OrderAmount_4 : 200

Memory_Address of OrderAmount_4 : 6487616

OrderAmount_5 : 421

Memory_Address of OrderAmount_5 : 6487620

-----
Process exited after 9.587 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

```
/*
    Create a function incrementFollowers(int *followers, int n)
    that increases each follower count in an array
    (representing Instagram followers for 5 friends)
    by 100 using pointer arithmetic, then print the updated counts.
*/

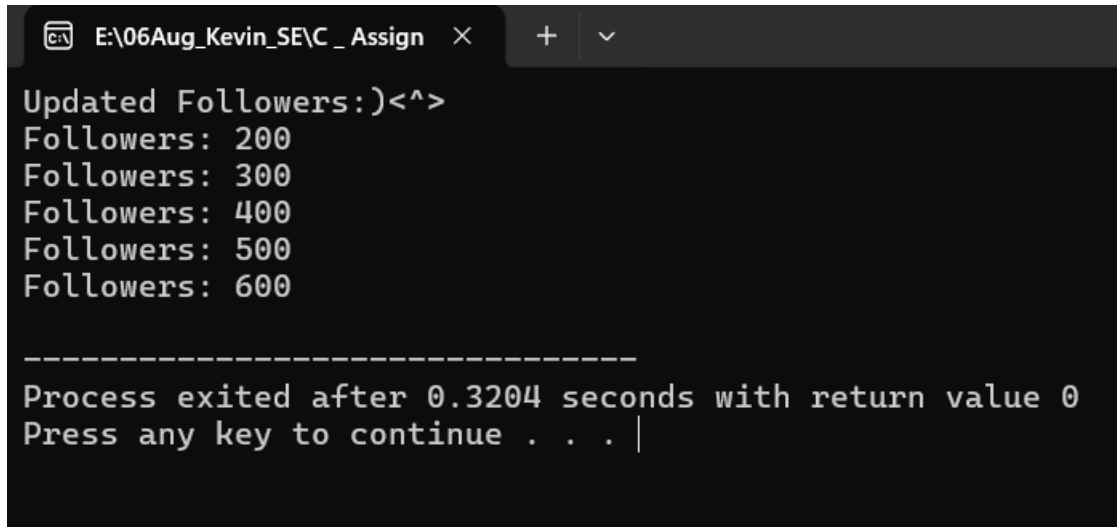
#include<stdio.h>
int increFollower(int *follower,int n)
{
    for(int i=0;i<n;i++)
    {
        *(follower+i) += 100;    // follower[i] += 100 alt;;;
    }
}
int main()
{
    int follower[5] = {100,200,300,400,500};

    increFollower(follower,5);

    printf("Updated Followers:<^>\n");

    for(int i=0;i<5;i++)
    {
        printf("Followers: %d\n",follower[i]);
    }
}
```

Output:



The screenshot shows a Windows command prompt window with a single tab titled "E:\06Aug_Kevin_SE\C_Assign". The output of the program is as follows:

```
Updated Followers:)<^>
Followers: 200
Followers: 300
Followers: 400
Followers: 500
Followers: 600

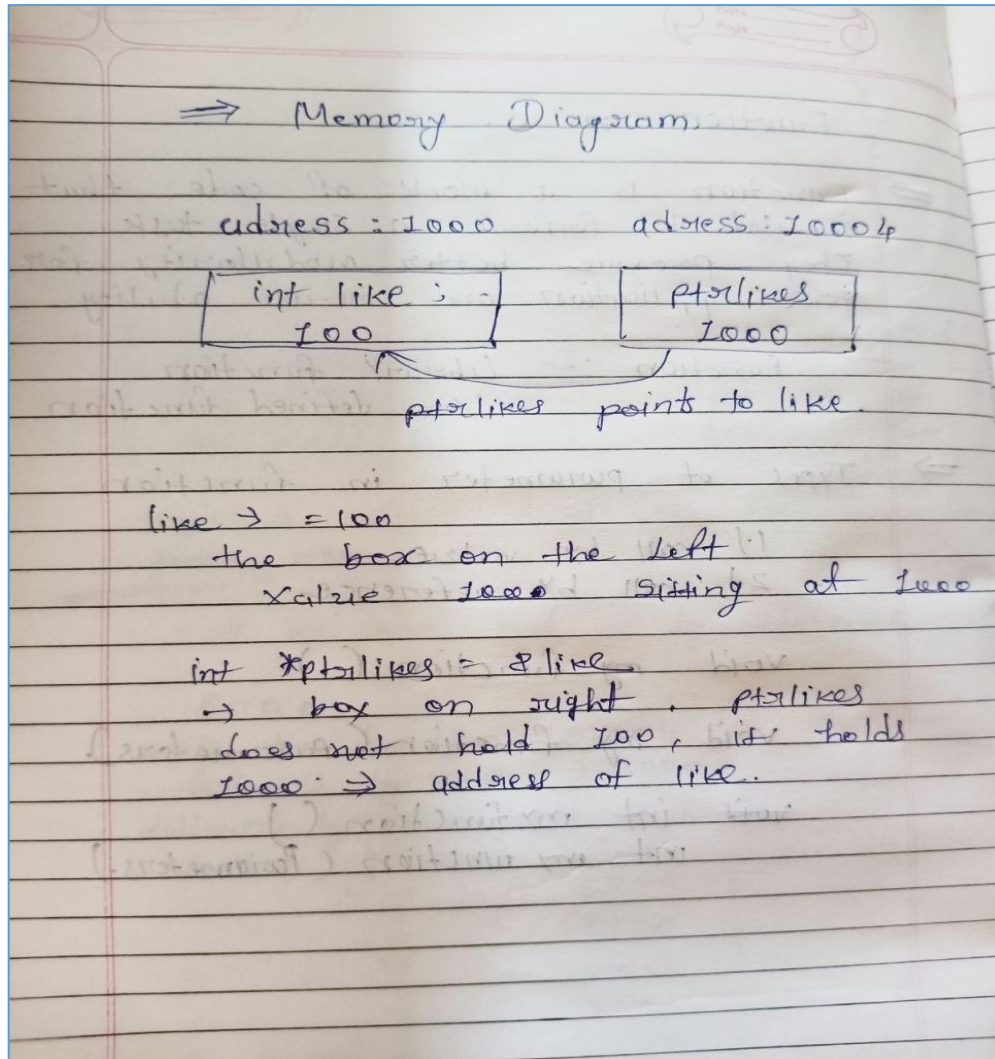
-----
Process exited after 0.3204 seconds with return value 0
Press any key to continue . . . |
```

Que - 5

/*

Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example; label the variable, pointer, and address.

*/



Session 12

Que – 1

```
/*
    Declare a structure named Playlist to store details of a song:
    title (string), artist (string),
    and duration in seconds (integer).
    Initialize one Playlist variable with your favorite song's details
    and print each field.
*/

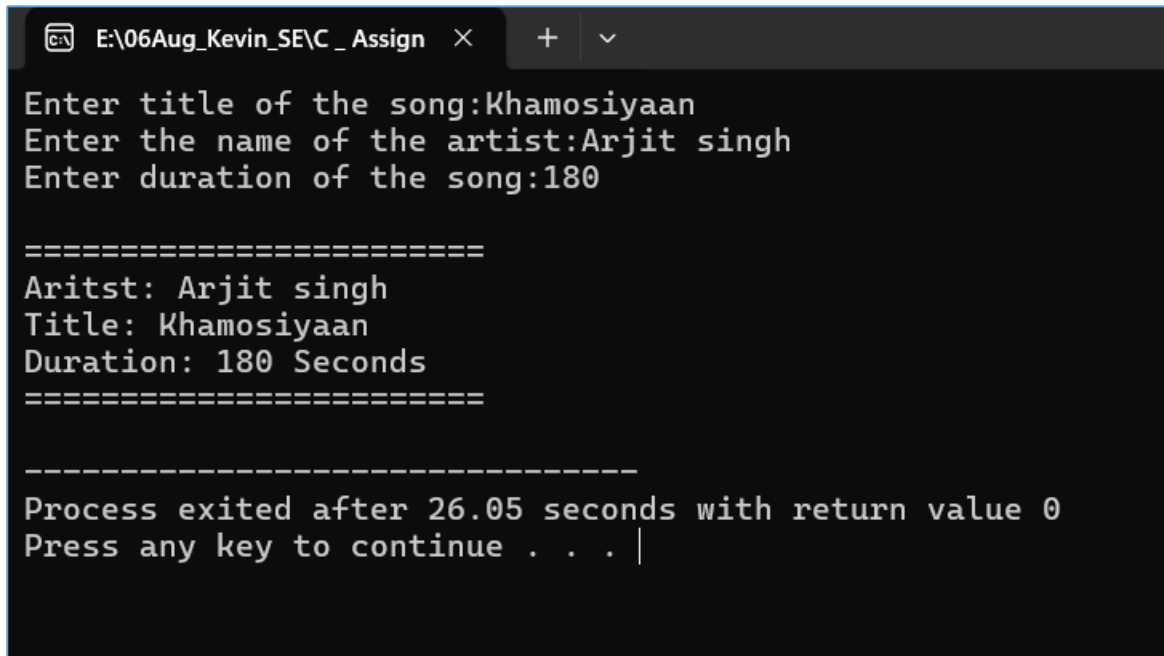
#include<stdio.h>

struct playList
{
    char title[50],artist[25];
    int duration;
} music;

int main()
{
    printf("Enter title of the song:");
    gets(music.title);
    printf("Enter the name of the artist:");
    gets(music.artist);
    printf("Enter duration of the song:");
    scanf("%d",&music.duration);

    printf("\n===== \n");
    printf("Artist: %s\n",music.artist);
    printf("Title: %s\n",music.title);
    printf("Duration: %d Seconds",music.duration);
    printf("\n===== \n");
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign × + ▾
Enter title of the song:Khamosiyaan
Enter the name of the artist:Arjit singh
Enter duration of the song:180

=====
Aritst: Arjit singh
Title: Khamosiyaan
Duration: 180 Seconds
=====

-----
Process exited after 26.05 seconds with return value 0
Press any key to continue . . . |
```

Que – 2

```
/*
    Create a structure called FoodItem to store Zomato-style menu
    data:
        itemName (string), price (float), and rating (float).
        Initialize an array of 3 FoodItem variables
        with real menu items and display their details using a loop.
*/
```

```
#include<stdio.h>
```

```
struct FoodItem
```

```
{
    char itemName[100];
    float price;
    float rating;
} zomato[20];
int main()
{
    int i,stock;
    printf("Enter the number of current stock:-");
    scanf("%d",&stock);
    for(i=0;i<stock;i++)
    {
        printf("Enter the name of available food at your Hotel:-");
        scanf("%s",zomato[i].itemName);

        printf("Enter price:-");
        scanf("%f",&zomato[i].price);

        printf("Enter ratings given by customer:-");
        scanf("%f",&zomato[i].rating);
    }

    printf("\n=====\\n");
    printf("\nFood Menu\\n");
    printf("\n=====\\n");
    for(i=0;i<stock;i++)
    {
        printf("Food Option_%d >>> %s\\n",i,zomato[i].itemName);
        printf("Price of Food Option_%d >>>
%.2f$\\n",i,zomato[i].price);
        printf("Rating of Food Option_%d >>>
%.1f\\n",i,zomato[i].rating);
        printf("\n=====\\n");
    }
}
```

```
    return 0;  
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign >
Enter the number of current stock:-2
Enter the name of available food at your Hotel:-Samosa
Enter price:-30
Enter ratings given by customer:-4.4
Enter the name of available food at your Hotel:-SandWich
Enter price:-50
Enter ratings given by customer:-5.5

=====

Food Menu

=====
Food Option_0 >>> Samosa
Price of Food Option_0 >>> 30.00$
Rating of Food Option_0 >>> 4.4

=====
Food Option_1 >>> SandWich
Price of Food Option_1 >>> 50.00$
Rating of Food Option_1 >>> 5.5

=====

-----
Process exited after 28.4 seconds with return value 0
Press any key to continue . . . |
```

Que – 3

```
/*
    Define a nested structure called MovieShow for a BookMyShow-style
    app:
    Movie (string), Screen (integer),
    and a nested structure Time
    with hours and minutes (integers).
    Create and initialize a MovieShow variable for any
    movie and print its details in the format
    'Movie: X, Screen: Y, Time: HH:MM'.
*/

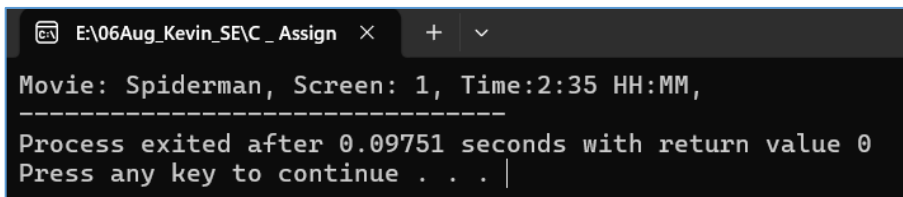
#include<stdio.h>

struct MovieShow
{
    char movie[15] = "Spiderman";
    int screen = 1;
};

struct movieLen
{
    int hour = 2;
    int min = 35;
    struct MovieShow obX; // nested structure with object
} obY; // object of 2nd struct

int main()
{
    printf("Movie: %s, Screen: %d, Time:%d:%d HH:MM,",
        obY.obX.movie,
        obY.obX.screen,
        obY.hour,
        obY.min);
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  × + ▾
Movie: Spiderman, Screen: 1, Time:2:35 HH:MM,
-----
Process exited after 0.09751 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

```
/*
    Build a structure called InstaProfile with fields:
    username (string),
    followers (integer),
    and a nested structure Bio with
    fields: description (string) and age (integer).
    Initialize an InstaProfile variable with
    your own details and display all fields.
*/

#include<stdio.h>

struct instaProfile
{
    char userName[25];
    int followers;
};

struct bio
{
    char description[100];
    int age;
    struct instaProfile pfp;
} likes;

int main()
{
    printf("Enter Your Username:");
    gets(likes.pfp.userName);

    printf("Enter Your Followers:");
    scanf("%d",&likes.pfp.followers);

    getchar();

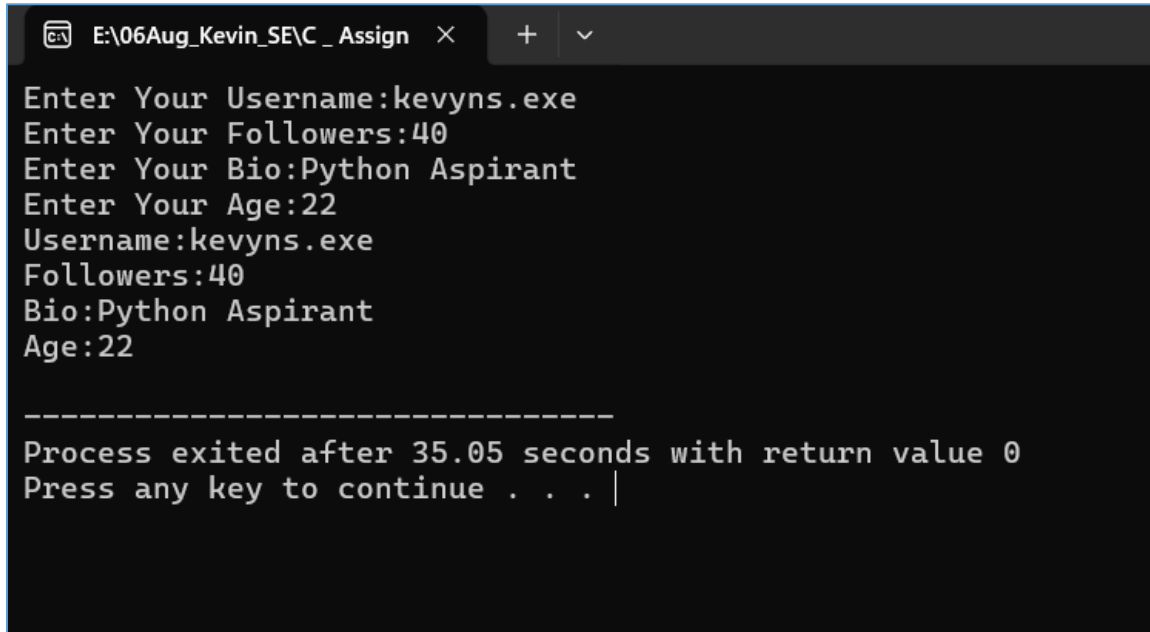
    printf("Enter Your Bio:");
    gets(likes.description);

    printf("Enter Your Age:");
    scanf("%d",&likes.age);

    printf("Username:%s\n",likes.pfp.userName);
    printf("Followers:%d\n",likes.pfp.followers);
}
```

```
    printf("Bio:%s\n",likes.description);  
    printf("Age:%d\n",likes.age);  
  
    return 0;  
}
```

Output:



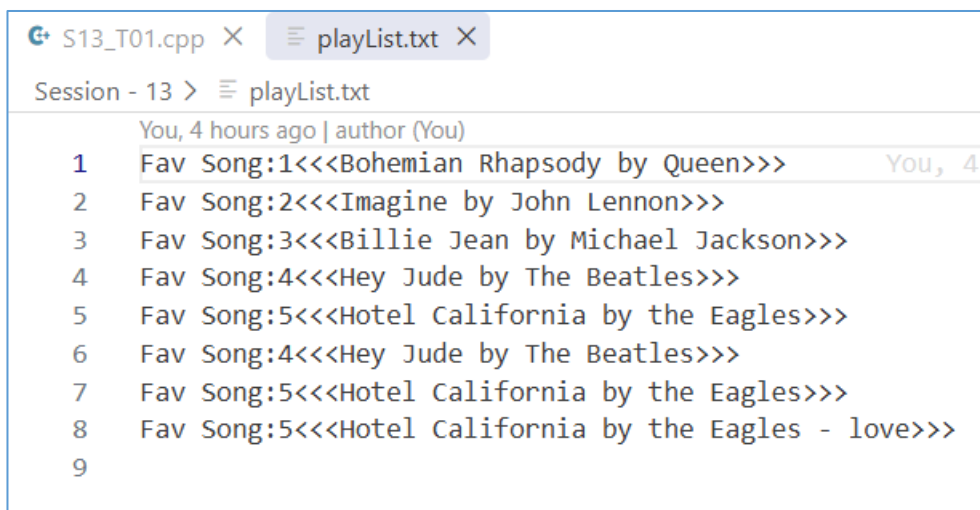
```
E:\06Aug_Kevin_SE\C_Assign  ×  +  ▾  
Enter Your Username:kevyns.exe  
Enter Your Followers:40  
Enter Your Bio:Python Aspirant  
Enter Your Age:22  
Username:kevyns.exe  
Followers:40  
Bio:Python Aspirant  
Age:22  
  
-----  
Process exited after 35.05 seconds with return value 0  
Press any key to continue . . . |
```

Session 13

Que – 1

```
/*  
    Create a file called  
    playlist.txt and write the names  
    of your top 3 favorite songs from  
    Spotify into it using write mode (w).  
*/  
  
#include<stdio.h>  
int main()  
{  
    FILE *ob;  
    ob = fopen("playList.txt","w");  
    //printf("-----File Created-----");  
    fprintf(ob,"Fav Song:1<<<Bohemian Rhapsody by Queen>>>\n");  
    fprintf(ob,"Fav Song:2<<<Imagine by John Lennon>>>\n");  
    fprintf(ob,"Fav Song:3<<<Billie Jean by Michael Jackson>>>\n");  
  
    fclose(ob);  
    printf("Yoo Done");  
}
```

Output:

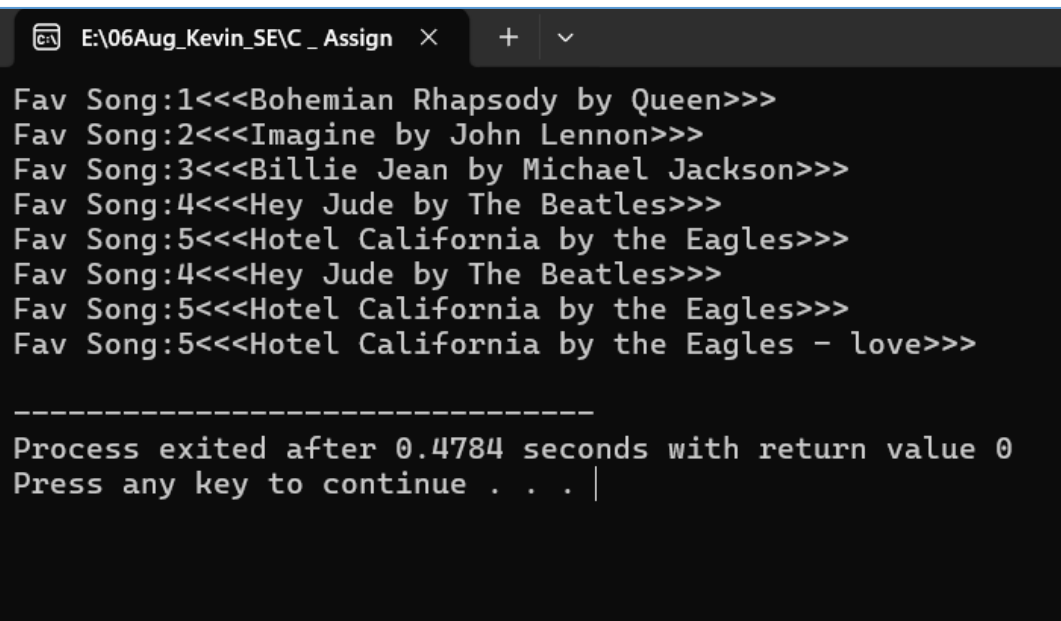


```
S13_T01.cpp X playList.txt X  
Session - 13 > playList.txt  
You, 4 hours ago | author (You)  
1 Fav Song:1<<<Bohemian Rhapsody by Queen>>> You, 4  
2 Fav Song:2<<<Imagine by John Lennon>>>  
3 Fav Song:3<<<Billie Jean by Michael Jackson>>>  
4 Fav Song:4<<<Hey Jude by The Beatles>>>  
5 Fav Song:5<<<Hotel California by the Eagles>>>  
6 Fav Song:4<<<Hey Jude by The Beatles>>>  
7 Fav Song:5<<<Hotel California by the Eagles>>>  
8 Fav Song:5<<<Hotel California by the Eagles - love>>>  
9
```

Que – 2

```
/*  
    Open playlist.txt in read mode (r) and  
    display each song name on a  
    separate line in the console.  
*/  
  
#include<stdio.h>  
int main()  
{  
    FILE *ob;  
    ob = fopen("playList.txt","r");  
  
    char fileData[100];  
    while(fgets(fileData,100,ob)!=NULL)  
    {  
        printf("%s",fileData);  
    }  
    fclose(ob);  
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  X  +  v  
Fav Song:1<<<Bohemian Rhapsody by Queen>>>  
Fav Song:2<<<Imagine by John Lennon>>>  
Fav Song:3<<<Billie Jean by Michael Jackson>>>  
Fav Song:4<<<Hey Jude by The Beatles>>>  
Fav Song:5<<<Hotel California by the Eagles>>>  
Fav Song:4<<<Hey Jude by The Beatles>>>  
Fav Song:5<<<Hotel California by the Eagles>>>  
Fav Song:5<<<Hotel California by the Eagles - love>>>  
  
-----  
Process exited after 0.4784 seconds with return value 0  
Press any key to continue . . . |
```

Que – 3

```

/*
    Add two more song names to playlist.txt
    without deleting the existing ones
    by opening the file in append mode (a).
*/

#include<stdio.h>
int main()
{
    FILE *ob;
    ob = fopen("playList.txt","a");

    fprintf(ob,"Fav Song:4<<<Hey Jude by The Beatles>>>\n");
    fprintf(ob,"Fav Song:5<<<Hotel California by the Eagles>>>\n");
    fprintf(ob,"Fav Song:5<<<Hotel California by the Eagles -
love>>>\n");

    printf("Yoo Done");

    return 0;
}

```

Output:

```

S13_T01.cpp X playList.txt X
Session - 13 > playList.txt
You, 4 hours ago | author (You)
1 Fav Song:1<<<Bohemian Rhapsody by Queen>>> You, 4
2 Fav Song:2<<<Imagine by John Lennon>>>
3 Fav Song:3<<<Billie Jean by Michael Jackson>>>
4 Fav Song:4<<<Hey Jude by The Beatles>>>
5 Fav Song:5<<<Hotel California by the Eagles>>>
6 Fav Song:4<<<Hey Jude by The Beatles>>>
7 Fav Song:5<<<Hotel California by the Eagles>>>
8 Fav Song:5<<<Hotel California by the Eagles - love>>>
9

```

Que – 4

```
/*
    Write a program that reads all song names
    from playlist.txt and prints only those
    that contain the word 'love' (case-insensitive).
    Hint:
    Use the 'in' keyword or equivalent string method for filtering.
*/

#include <stdio.h>
#include <string.h>
#include <ctype.h>

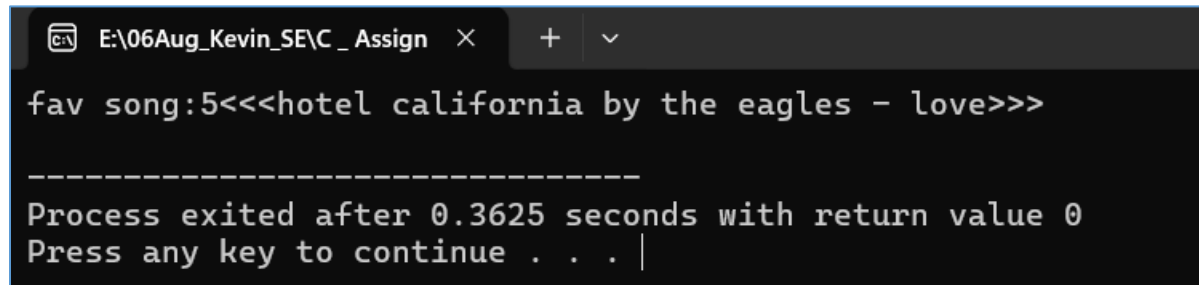
int main()
{
    FILE *ob;
    char song[100];

    ob = fopen("playList.txt", "r");

    while (fgets(song, 100, ob))
    {
        for (int i = 0; song[i] != '\0'; i++)
        {
            song[i] = tolower(song[i]);
        }

        if (strstr(song, "love") != NULL)
        {
            printf("%s", song);
        }
    }
    fclose(ob);
    return 0;
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign × + ∨  
fav song:5<<<hotel california by the eagles - love>>>  
-----  
Process exited after 0.3625 seconds with return value 0  
Press any key to continue . . . |
```

Session 14

Que -1

```
/*  
    Given the following buggy code meant to  
    calculate the total price of a Zomato order,  
    identify and fix the syntax and runtime errors:
```

```
    let items = ["Burger", "Pizza", "Fries"];  
    let prices = [120, 250, 90];  
    let total = 0;  
    for (i = 0; i < items.length; i++)  
    {  
        total += prices[i]<br>  
        console.log("Total price is: " + total);
```

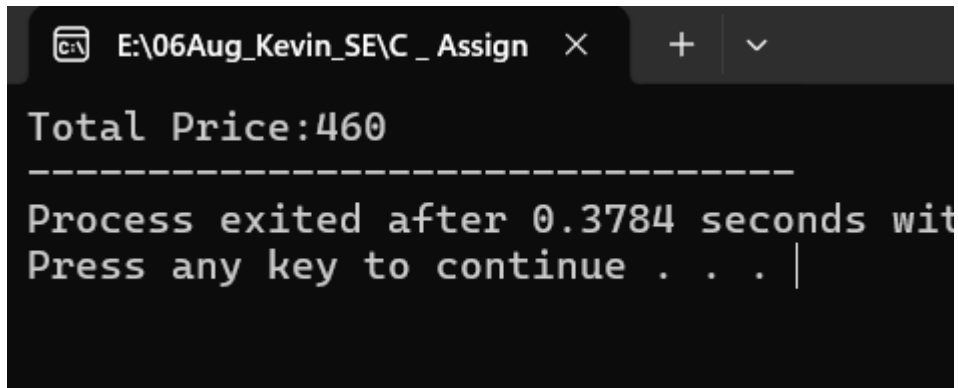
Hint:

Watch for assignment and loop variable issues.

```
*/
```

```
#include<stdio.h>  
int main()  
{  
    char items[10][10] = {"Burger","Pizza","Fries"};  
    int prices[10] = {120,250,90};  
    int total,i;  
  
    total =0;  
  
    for(i=0;i<3;i++)  
    {  
        total = total + prices[i];  
    }  
    printf("Total Price:%d",total);  
    return 0;  
}
```

Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\06Aug_Kevin_SE\C_Assign' and standard window controls. The command prompt displays the output of a program: 'Total Price:460' followed by a line of dashes, then 'Process exited after 0.3784 seconds with', and finally 'Press any key to continue . . . |' with a cursor at the end.

```
E:\06Aug_Kevin_SE\C_Assign >
Total Price:460
-----
Process exited after 0.3784 seconds with
Press any key to continue . . . |
```

Que – 2

/*
Rewrite the following code to improve its indentation
and add comments explaining each step,
so that a beginner can understand what it does:

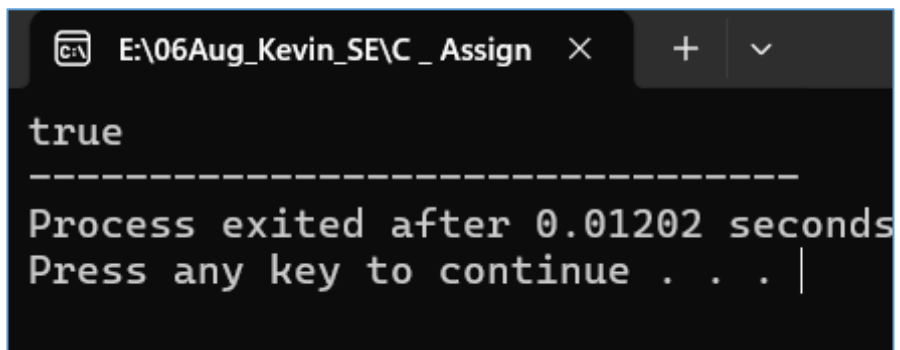
```
function isEven(num)
{
    if(num%2==0)
    {
        return true;
    }
    else{return false;
}
*/

#include<stdio.h>
#include<stdbool.h>

bool isEven(int num)
{
    if(num%2 == 0 )
    {
        return true;
    }
    else
    {
        return false;
    }
}

int main()
{
    int number;
    number = 204;
    bool result;
    result = isEven(number);
    printf("%s",result ? "true" : "false");
    return 0;
}
```

Output:



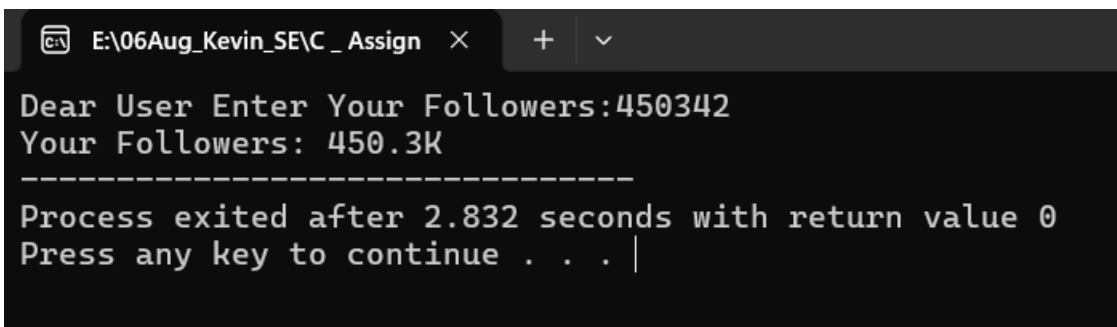
```
E:\06Aug_Kevin_SE\C_Assign  X  +  v
true
-----
Process exited after 0.01202 seconds
Press any key to continue . . . |
```

Que – 3

```
/*
Write a function formatFollowersCount(count)
that takes a number and returns a formatted string like Instagram:
1500 as '1.5K',
1200000 as '1.2M',
and numbers below
1000 as-is.
Add clear comments and use proper indentation.
*/
#include<stdio.h>
int main()
{
    float i;
    printf("Dear User Enter Your Followers:");
    scanf("%f",&i);

    if(i>1000000)
    {
        printf("Your Followers: %.1fM",i/1000000.0);
    }
    else if(i>=1000)
    {
        printf("Your Followers: %.1fK",i/1000.0);
    }
    else
    {
        printf("Your Followers: %.0f",i);
    }
}
```

Output:



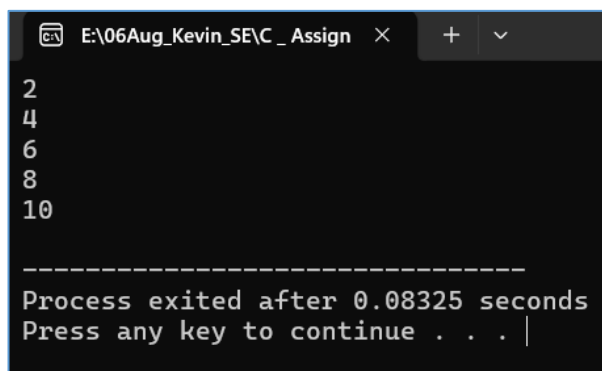
```
E:\06Aug_Kevin_SE\C_Assign
Dear User Enter Your Followers:450342
Your Followers: 450.3K
-----
Process exited after 2.832 seconds with return value 0
Press any key to continue . . . |
```

Que – 4

```
/*  
    You are given a piece of JavaScript code  
    that should print all even numbers from  
    1 to 10, but it doesn't work as expected:  
  
    for (let i = 1; i <= 10; i++)  
    {  
        if (i % 2 = 0)  
        {  
            console.log(i)  
        }  
    }  
    Find and fix the error,  
    then rewrite the code with good indentation  
    and at least one comment explaining the logic.  
*/
```

```
#include<stdio.h>  
int main()  
{  
    int i;  
    for (i=1;i<=10;i++)  
    {  
        if(i%2==0) // Operator Error was present in javascript code  
        {  
            printf("%d\n",i);  
        }  
    }  
}
```

Output:



```
E:\06Aug_Kevin_SE\C_Assign  X  +  v  
2  
4  
6  
8  
10  
-----  
Process exited after 0.08325 seconds  
Press any key to continue . . . |
```